

ETL Pipeline

TA. Đăng Nhã
STA. Anh Dương

Outline

Overview

1. Data-driven decision making
2. Business Intelligent
3. Data Architecture
4. ETL process overview

Section 1

Data Transformation

1. Transformation Problem
2. Cleaning
3. Processing

Section 3

Data Extraction

1. Extraction Process
2. Dataset: Iowa Liquor Sale
3. Case study: Extract Iowa Liquor Sale Dataset

Section 2

Data Loading

1. Dims, Fact
2. Slowly Changing Dimension
3. Full Pipeline

Section 4

[Quiz&Feedbacks](#)

Overview



Overview

❖ Problem



Collect Data



Company Departments

Marketing

Operations

Financial

Reporting



Transactions Inventory Purchasing

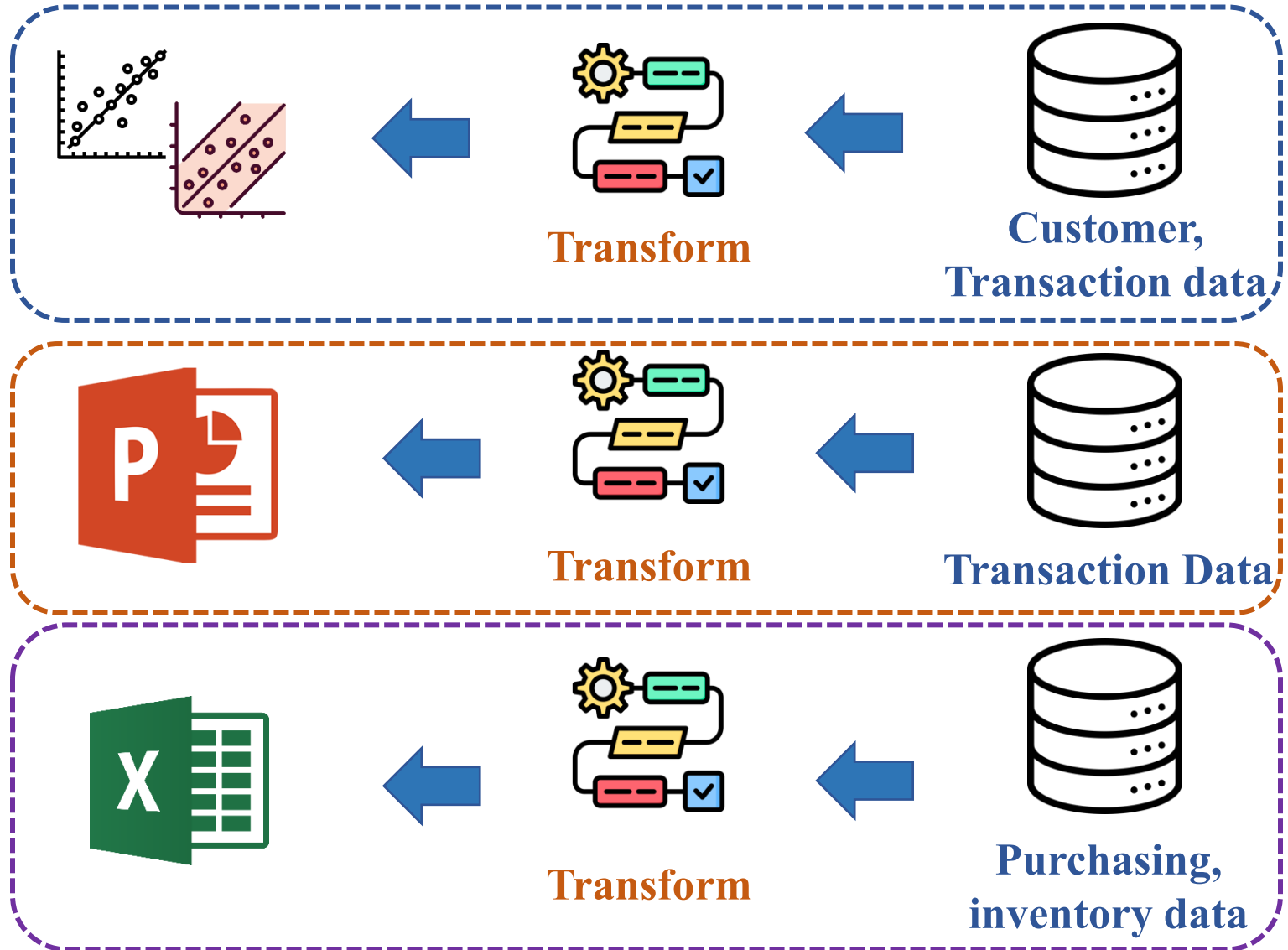


Retail Company

Overview

❖ Why we need report?

- What was the current trends in customer behavior in the past 2 years?
- What was our total retail sales last month?
- What was the amount of inventory of key products over the last three months



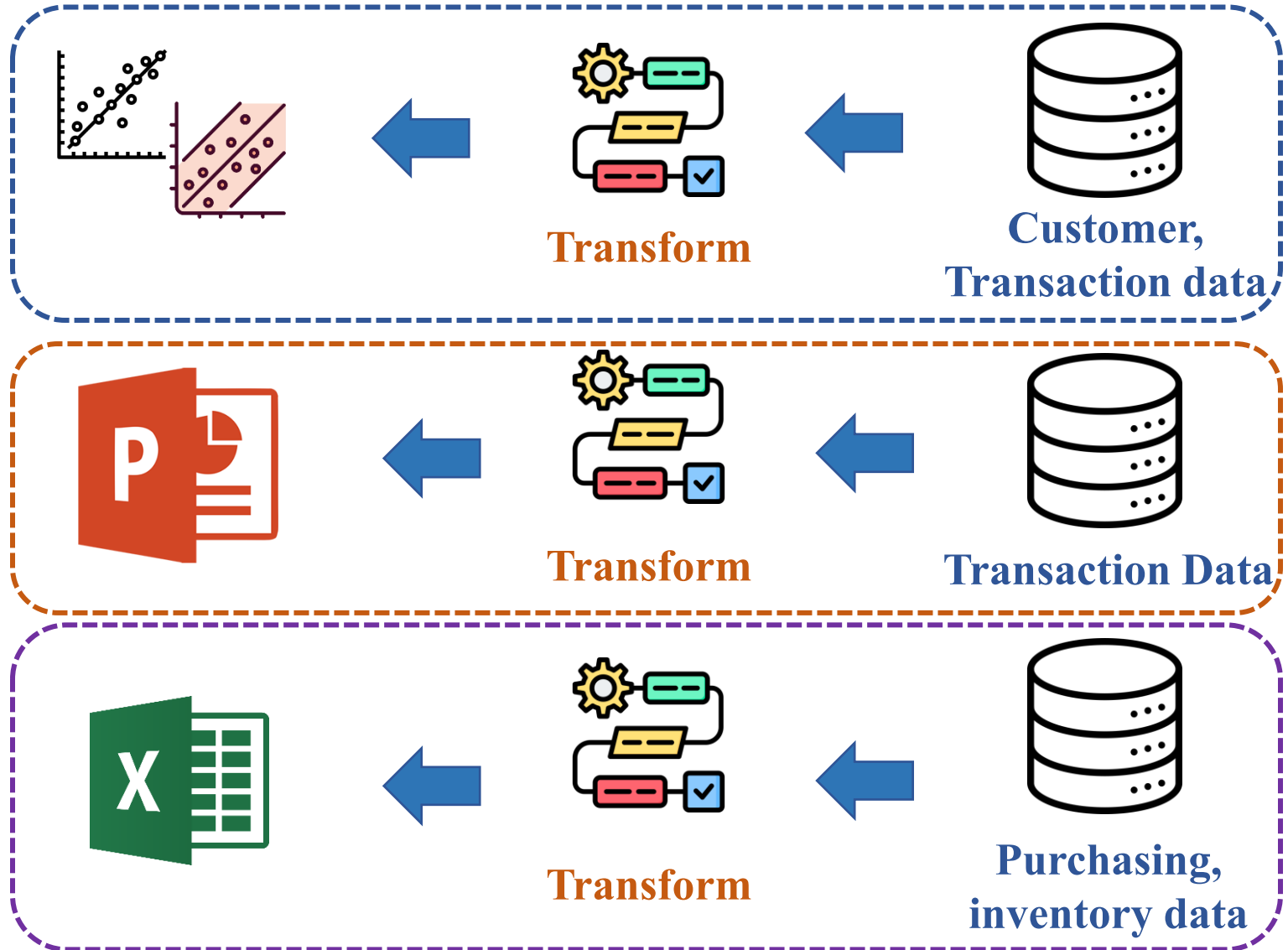
Overview

❖ Why we need report?

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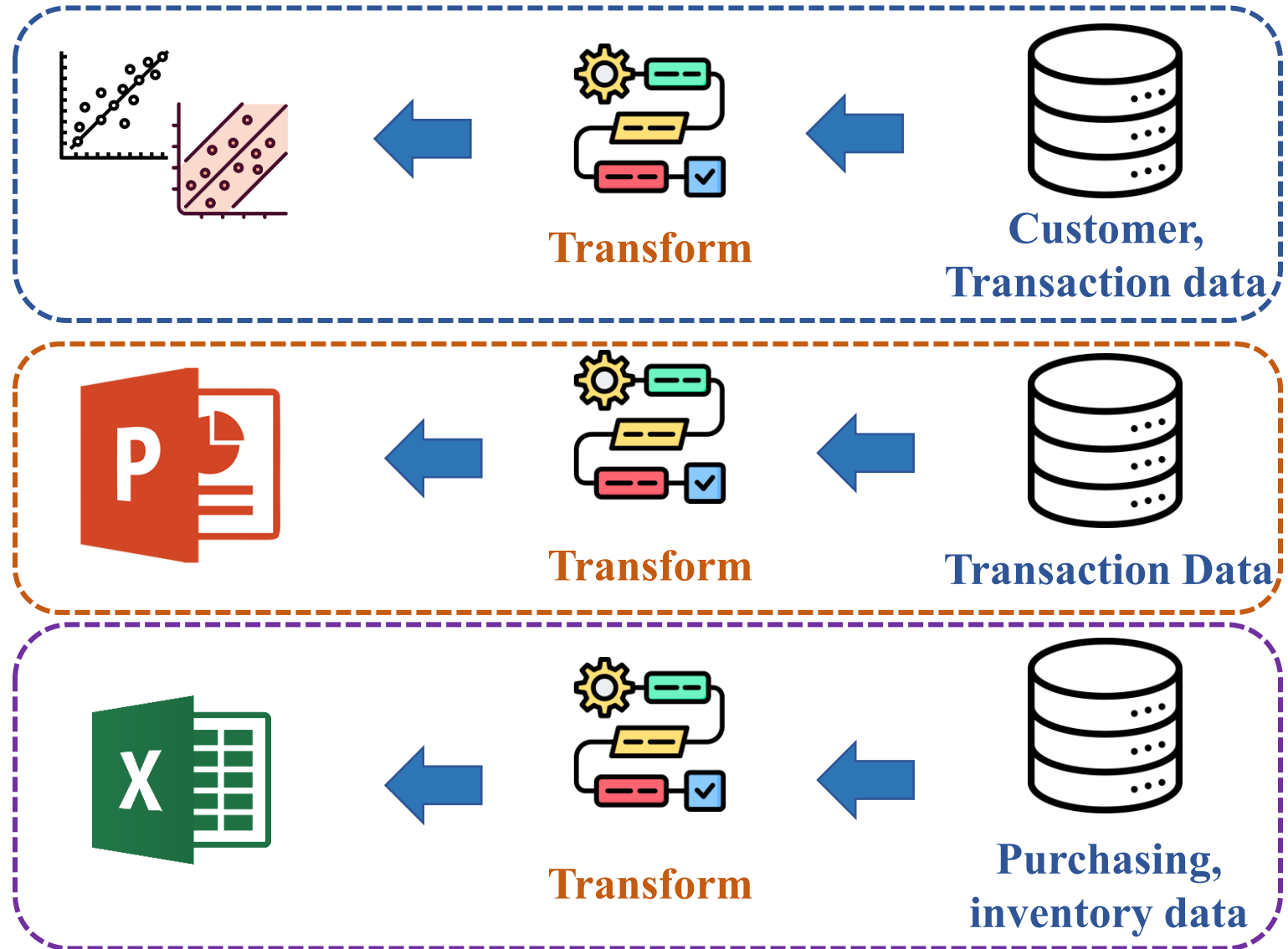
Problem?



Overview

❖ Why we need report?

- Time Consuming (Build report from scratch, collect data from many source)
- Many human involve (inconsistence, human error)
- Hard to have intergrated report (many tools, many terminologies,)
- Unpredicted Issues (number not match, wrong information)





Overview

❖ Solutions



“We need a centralized, reliable, and scalable way to use our data.”



Data
Warehouse

Structure Data



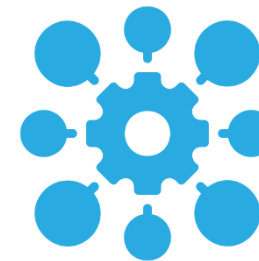
Data
Lake

Semi, Unstructure
Data



Data
Lakehouse

Mix Warehouse
and Lake



Data
Mesh

Decentralize



Overview

❖ Solutions



“We need a centralized, reliable, and scalable way to use our data.”



Data Warehouse

Structure Data



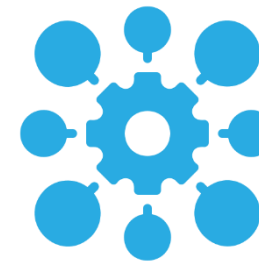
Data Lake

Semi, Unstructure Data



Data Lakehouse

Mix Warehouse and Lake



Data Mesh

Decentralize



Overview

❖ Data Warehouse – As a Restaurant

Multi Source Ingredients

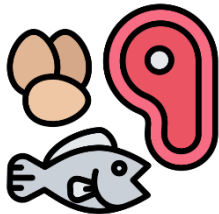
Market



Mart



Farm



Warehouse

Collect

Cleaning

Processing

Preserve

Arrange

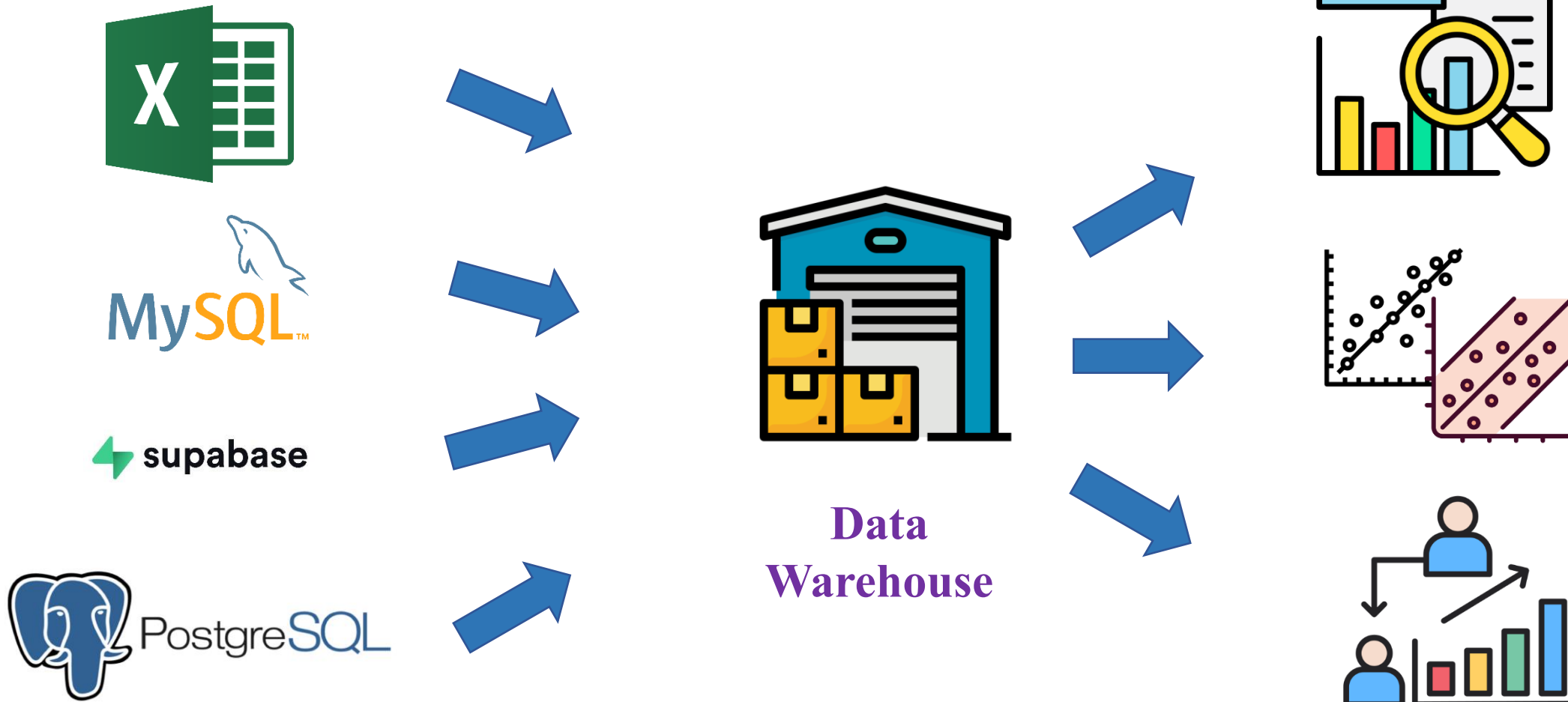
Kitchen

Choose Processed
Ingredients

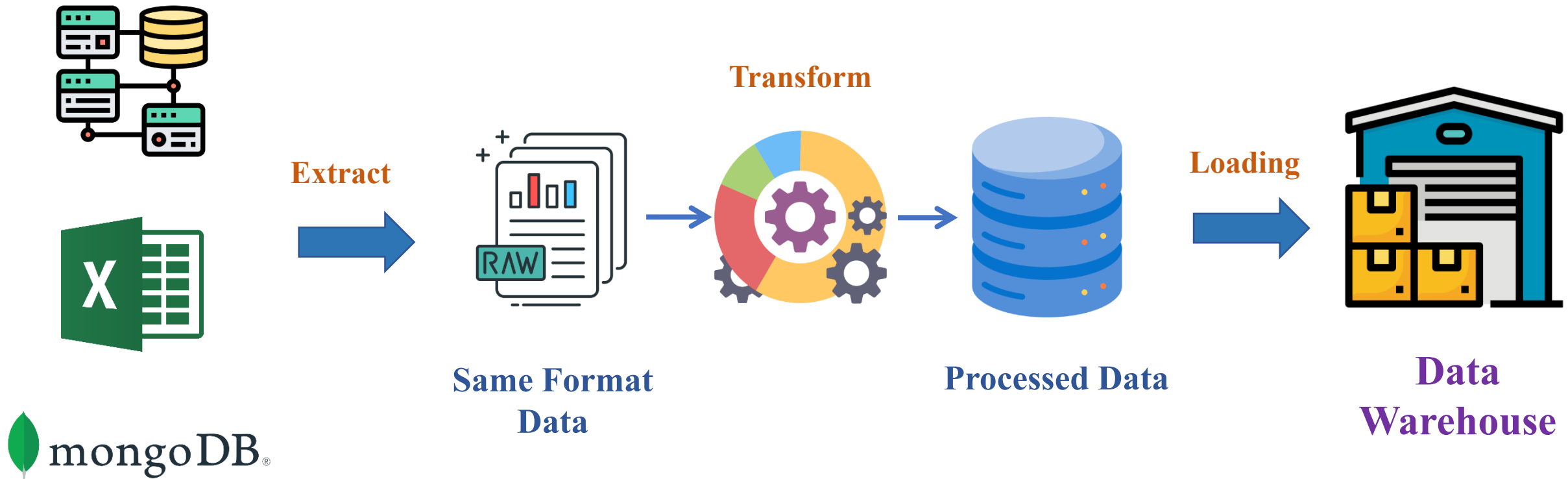
Cooking,
Decorating

Serving Customer

❖ Data Warehouse – As a Restaurant



❖ ETL process

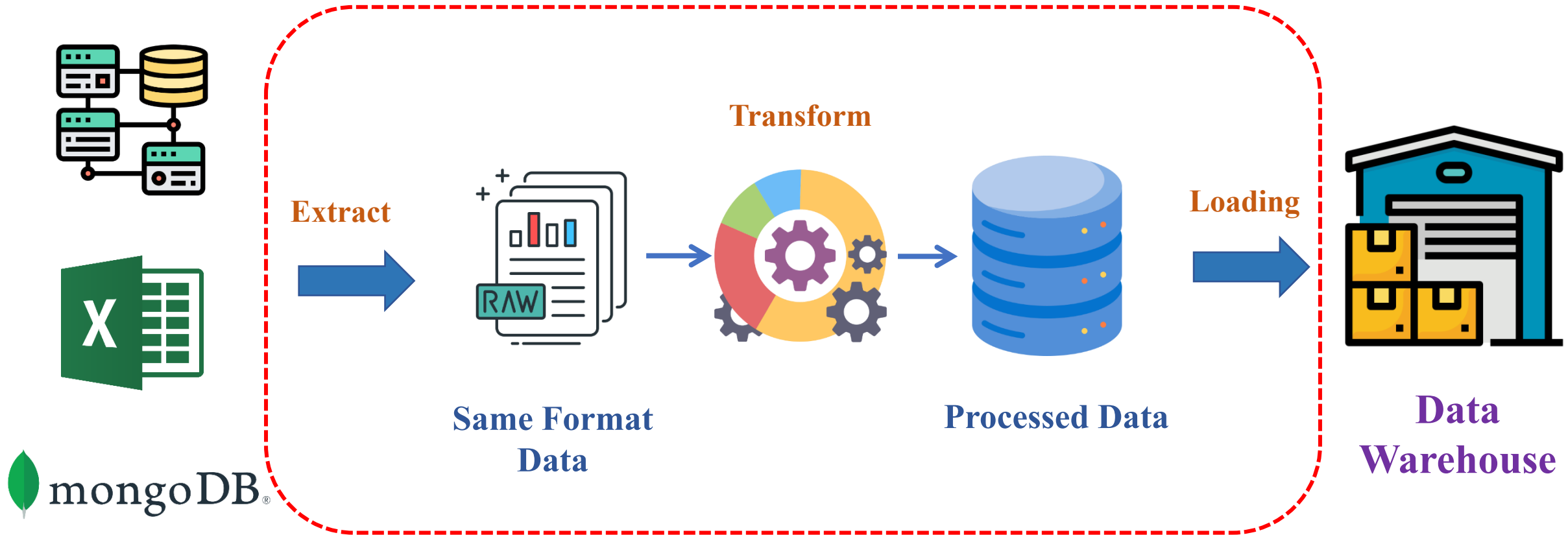


Multi Sources



Overview

❖ ETL process



Multi Sources

❖ Data-driven Decision Making

“Data-driven decision-making (DDDM) is an approach that emphasizes using data and analysis instead of intuition to inform business decisions.”

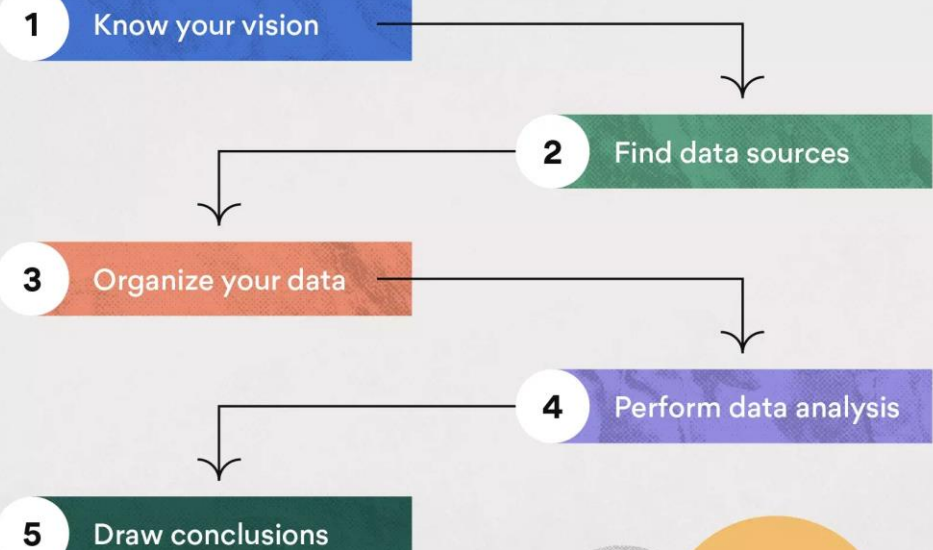


Intuitive



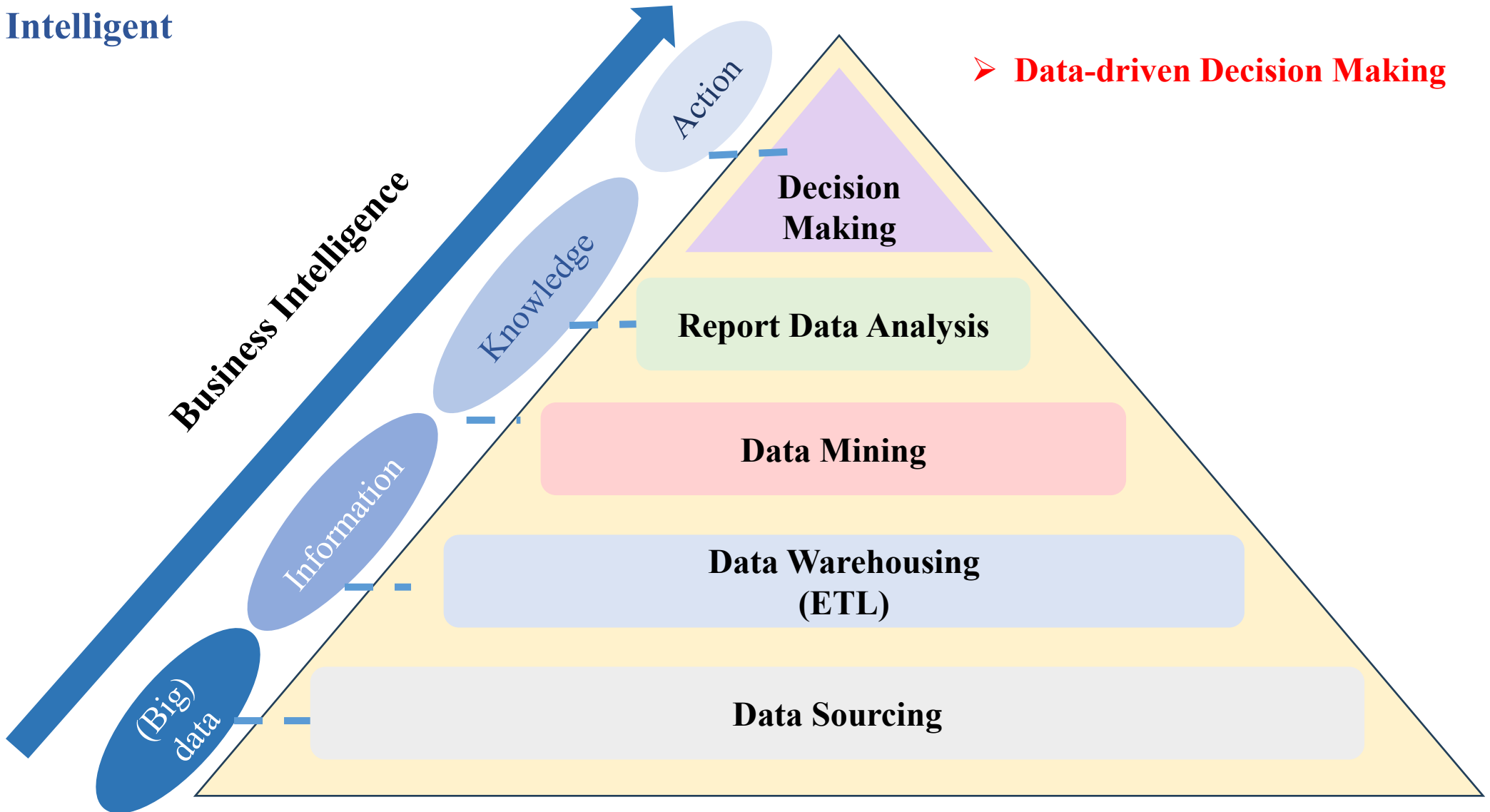
Quantitative

How to make data-driven decisions



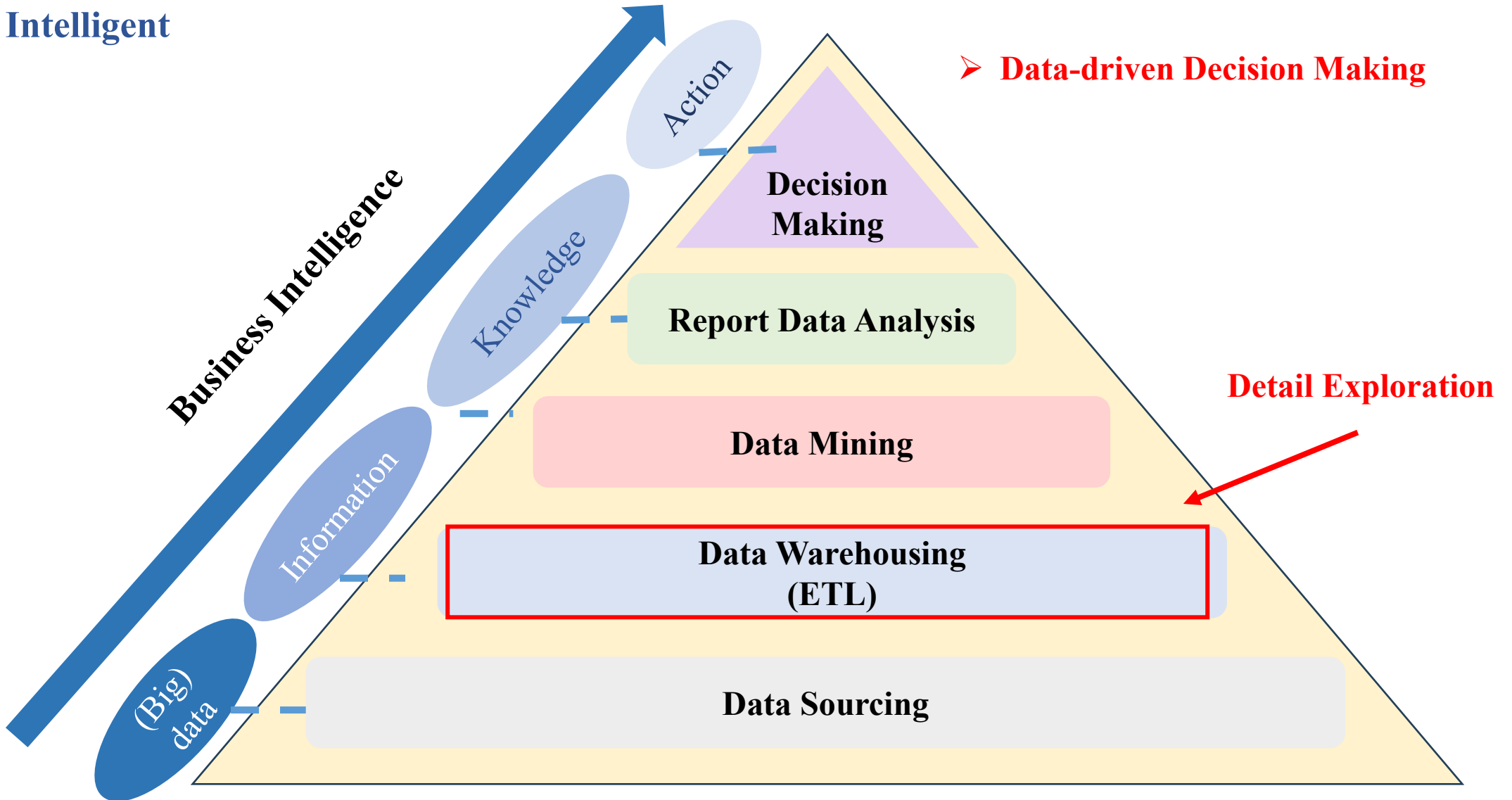
Overview

❖ Business Intelligent



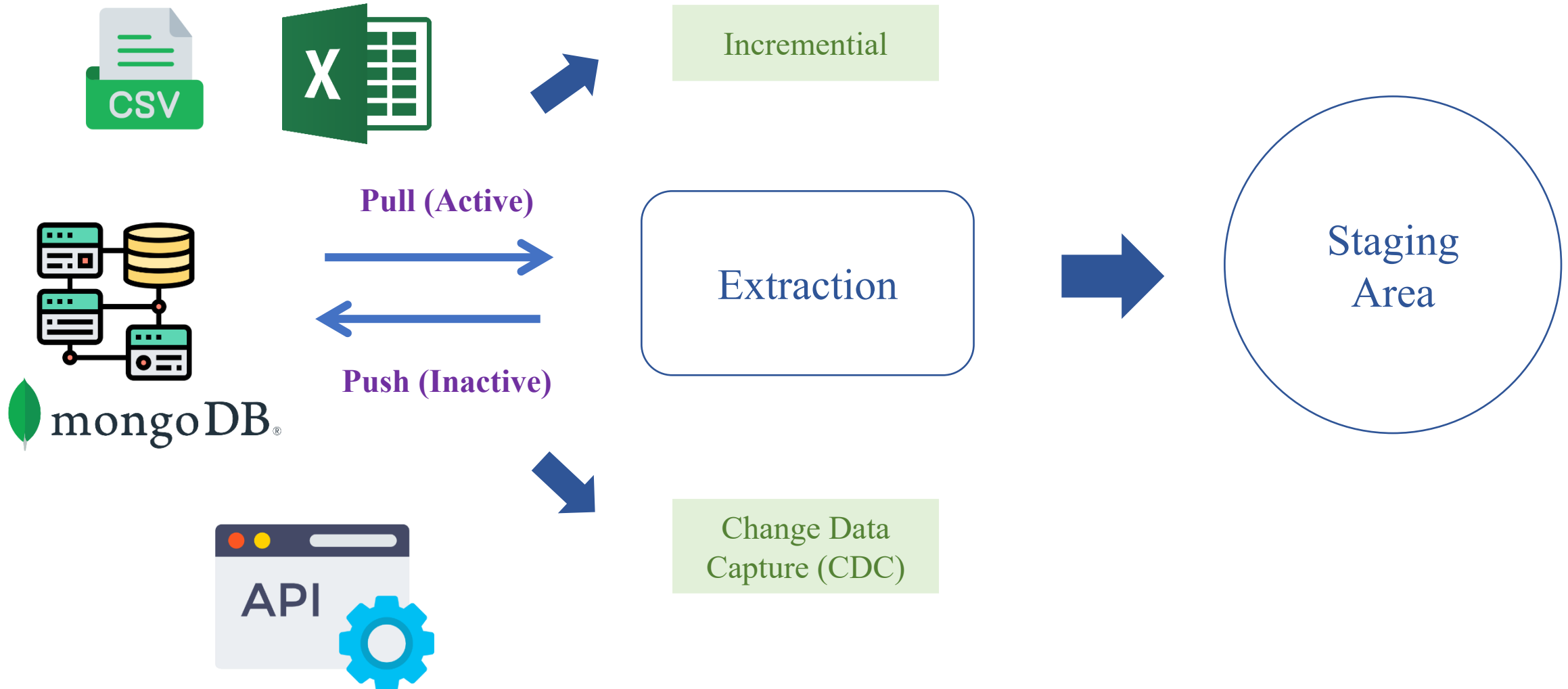
Overview

❖ Business Intelligent



Data Extraction

❖ What is Data Extraction?





Data Extraction

❖ Problem during Extraction Process

Store A			
store_id	transac_id	amount	cost
st00a	tran001	1	2
st00a	tran002	3	2.3
st00a	tran003	2	1.9

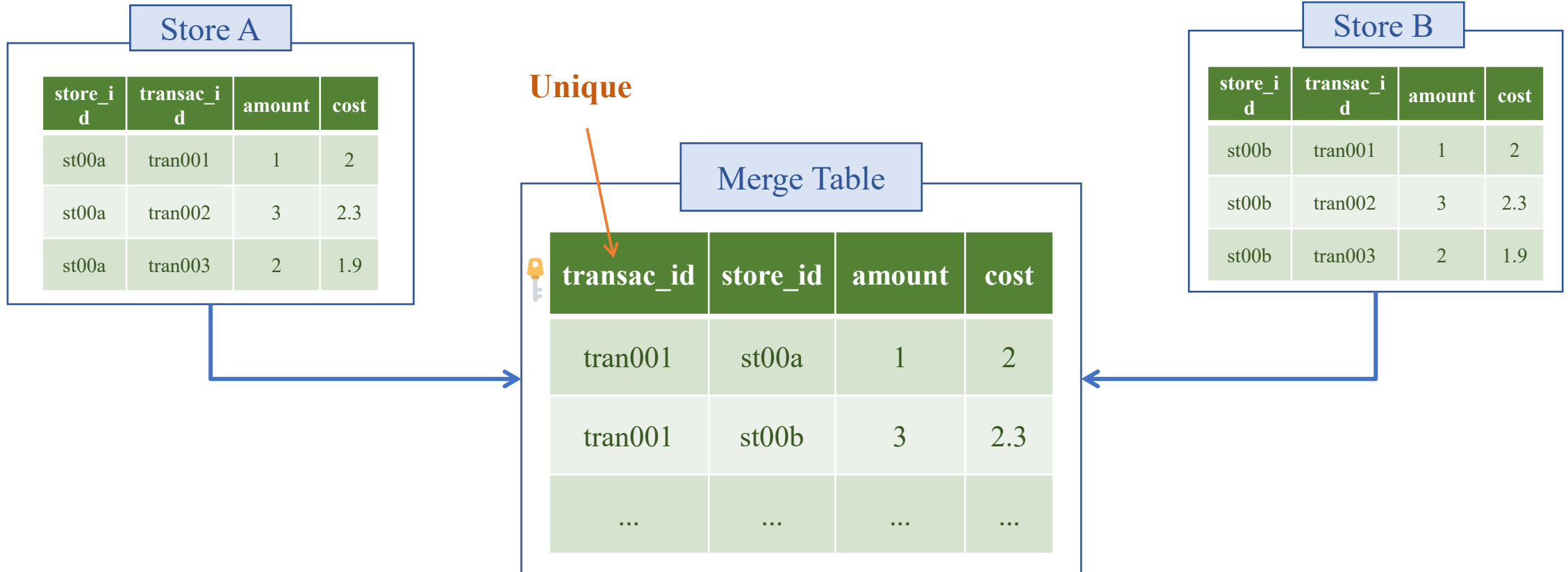
Store B			
store_id	transac_id	amount	cost
st00b	tran001	1	2
st00b	tran002	3	2.3
st00b	tran003	2	1.9

Composite Key Collisions



Data Extraction

❖ Problem during Extraction Process

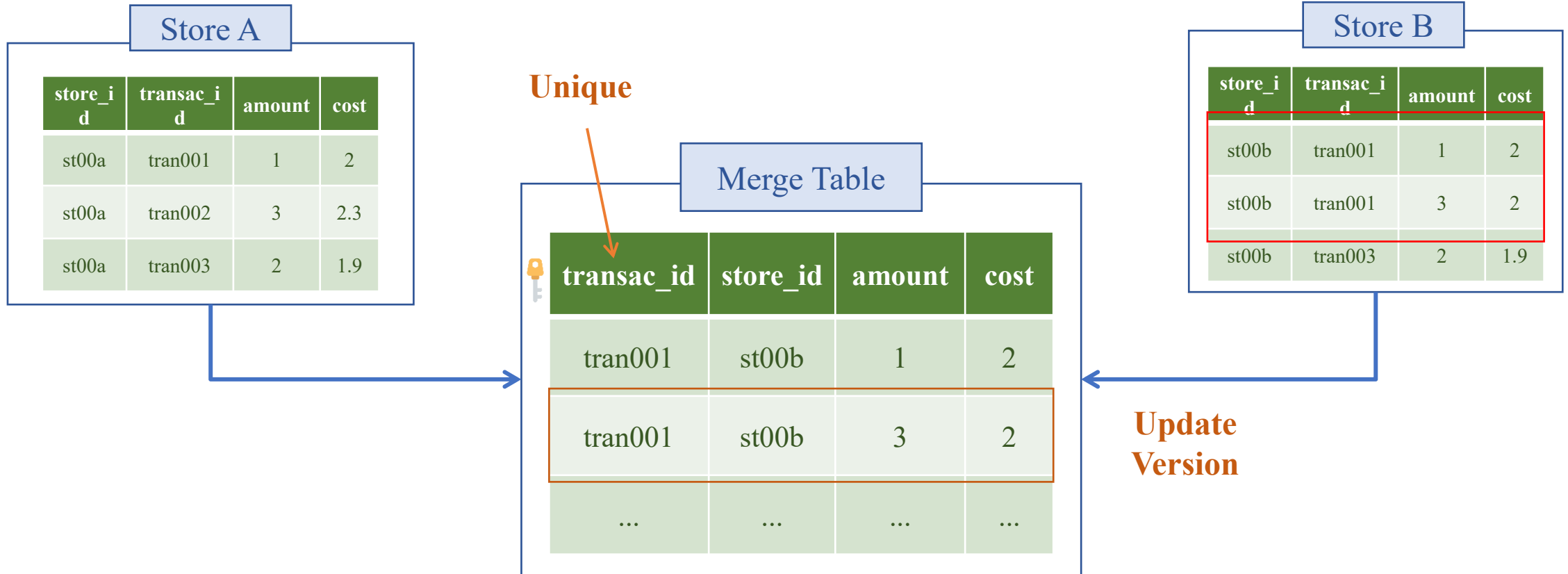


Composite Key Collisions



Data Extraction

❖ Problem during Extraction Process



Tailored Storing Method

❖ Case Study: Iowa Liquor Sale

 UMAIR ZIA · UPDATED 2 MONTHS AGO

▲ 1

<> Code

Download

Iowa Liquor Sales

Can You Identify Liquor Trends??

Data Card

Code (0)

Discussion (0)

Suggestions (0)

About Dataset

This dataset contains the spirits purchase information of Iowa Class "E" liquor licensees by product and date of purchase from January 1, 2012 to current. The dataset can be used to analyze total spirits sales in Iowa of individual products at the store level.

Class E liquor license, for grocery stores, liquor stores, convenience stores, etc., allows commercial establishments to sell liquor for off-premises consumption in original unopened containers.

Source: [Link](#)

License: CC0 1.0 Universal

Below are the main columns affecting sales

Column Name	Description	API Field Name	Data Type
-------------	-------------	----------------	-----------

Usability ⓘ
9.41

License
Other (specified in description)

Expected update frequency
Annually

Tags
Alcohol

Iowa Liquor Sales

❖ Case Study: Iowa Liquor Sale

```
RangeIndex: 1000000 entries, 0 to 999999
Data columns (total 24 columns):
#   Column                Non-Null Count  Dtype
---  -
0   invoice_line_no       1000000 non-null object
1   date                  1000000 non-null object
2   store                 1000000 non-null int64
3   name                  1000000 non-null object
4   address               993904 non-null object
5   city                  993903 non-null object
6   zipcode               993912 non-null object
7   store_location        919061 non-null object
8   county_number         991839 non-null object
9   county                993481 non-null object
10  category               998518 non-null object
11  category_name          998563 non-null object
12  vendor_no              999999 non-null object
13  vendor_name            999999 non-null object
14  itemno                 1000000 non-null object
15  im_desc                1000000 non-null object
16  pack                   1000000 non-null float64
17  bottle_volume_ml       1000000 non-null float64
18  state_bottle_cost      999826 non-null object
19  state_bottle_retail    999826 non-null object
20  sale_bottles            999826 non-null object
21  sale_dollars            999060 non-null object
22  sale_liters             999272 non-null object
23  sale_gallons            990034 non-null object
dtypes: float64(2), int64(1), object(21)
memory usage: 183.1+ MB
```

Column Name	Description
Invoice/Item Number	Concatenated invoice and line number associated with the liquor order. Provides a unique identifier for each liquor product in the store order.
Date	Date of order
Store Number	Unique number assigned to the store who ordered the liquor.
Store Name	Name of store who ordered the liquor.
Address	Address of store who ordered the liquor.
City	City where the store is located.
Zip Code	Zip code of the store location.
Store Location	Geographic coordinates from geocoding the store's address, city, and zip code. NULL if unable to geocode.
County Number	Iowa county number where the store is located.
County	Name of the county where the store is located.
Category	Category code of the liquor ordered.
Category Name	Category name of the liquor ordered.

Vendor Number	Vendor number of the liquor brand.
Vendor Name	Vendor name of the liquor brand.
Item Number	Item number for the individual liquor product.
Item Description	Description of the individual liquor product.
Pack	Number of bottles per case.
Bottle Volume (ml)	Volume of each bottle in milliliters.
State Bottle Cost	Price paid by the Alcoholic Beverages Division per bottle.
State Bottle Retail	Price paid by the store per bottle.
Bottles Sold	Number of bottles sold in the order.
Sale (Dollars)	Total cost of the order: <code>state_bottle_retail * sale_bottles</code> .
Volume Sold (Liters)	Total volume sold in liters: <code>(bottle_volume_ml * sale_bottles) / 1000</code> .
Volume Sold (Gallons)	Total volume sold in gallons: <code>(bottle_volume_ml * sale_bottles) / 3785.411784</code> .



Data Extraction

❖ Case Study: Iowa Liquor Sale



**Extract
Data**

**Data
Cleaning**



Duplicated

Missing

Type
Casting

Filtering

Invalid

**Data
Processing**

Enrichment

Aggregate

Derived
Column

Normalize

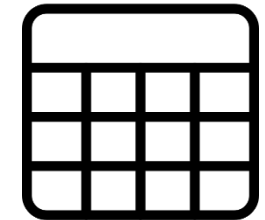
Intergrate

**Transform
Data**

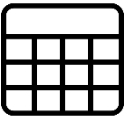
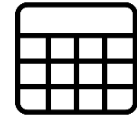
Warehouse



Fact



dims



**Load
Data**

❖ Case Study: Iowa Liquor Sale





❖ Case Study: Iowa Liquor Sale



100k rows

Batch
Processing



10k rows



```
DROP TABLE IF EXISTS Staging_Sales;  
CREATE TABLE Staging_Sales (  
    invoice_line_no TEXT,  
    date TEXT,  
    store TEXT,  
    name TEXT,  
    address TEXT,  
    city TEXT,  
    zipcode TEXT,  
    store_location TEXT,  
    county_number TEXT,  
    county TEXT,  
    category TEXT,  
    category_name TEXT,  
    vendor_no TEXT,  
    vendor_name TEXT,  
    itemno TEXT,  
    im_desc TEXT,  
    pack TEXT,  
    bottle_volume_ml TEXT,  
    state_bottle_cost TEXT,  
    state_bottle_retail TEXT,  
    sale_bottles TEXT,  
    sale_dollars TEXT,  
    sale_liters TEXT,  
    sale_gallons TEXT,  
    file_name TEXT NOT NULL,  
    processed_timestamp DATETIME NOT NULL,  
);
```



Data Extraction

❖ Case Study: Iowa Liquor Sale

New File



Batch
Processing



10k rows



Processed



```
CREATE TABLE Staging_Sales (  
    invoice_line_no TEXT,  
    date TEXT,  
    store TEXT,  
    name TEXT,  
    address TEXT,  
    city TEXT,  
    zipcode TEXT,  
    store_location TEXT,  
    county_number TEXT,  
    county TEXT,  
    category TEXT,  
    category_name TEXT,  
    vendor_no TEXT,  
    vendor_name TEXT,  
    itemno TEXT,  
    im_desc TEXT,  
    pack TEXT,  
    bottle_volume_ml TEXT,  
    state_bottle_cost TEXT,  
    state_bottle_retail TEXT,  
    sale_bottles TEXT,  
    sale_dollars TEXT,  
    sale_liters TEXT,  
    sale_gallons TEXT,  
    file_name TEXT NOT NULL,  
    processed_timestamp DATETIME NOT NULL,  
    record_source TEXT NOT NULL  
);
```

❖ Case Study: Iowa Liquor Sale

SQL ▼	< 1 / 2000 > 1 - 50 of 10000														🔍 🔄 📄	
invoice_line_no	date	store	name	address	city	zipcode	store_location	county_number	county	category	category_name	vendor_no	vendor_name	itemno		
506286000031	2012-06-27T00:00:00.000	3615	EMWETSBURG FOOD PRIDE	1307 S BROADWAY	EMWETSBURG	50536	74	PALO ALTO	1001330	PEACH SCHWAPPS	434	LUXCO-ST LOUIS	84456	PARAMOUNT PEACH		
516954500001	2014-01-22T00:00:00.000	4163	FAREWAY STORES #461 / STORM LAKE	731 SENECA ST	STORM LAKE	50588	11	BUENA VISTA	1011100	BLENDED WHISKIES	260	DIAGEO AMERICAS	25606	SEAGRAMS 7 CROWN		
507013300025	2012-08-08T00:00:00.000	4052	MELLE HOWETOWN VARIETY	300 1ST ST, STE F	SERGEANT BLUFF	51054	97	WOODBURY	1012200	SCOTCH WHISKIES	370	PERNOD RICARD USA/AUSTIN NICHOLS	9428	PASSPORT SCOTCH		
510581800183	2013-02-13T00:00:00.000	2501	HY-VEE #2 / AMES	640 LINCOLNWAY	AMES	50010	85	STORY	1012210	SINGLE MALT SCOTCH	370	PERNOD RICARD USA/AUSTIN NICHOLS	5037	GLENLIVET 12 YR F		
504395200026	2012-03-05T00:00:00.000	2505	HY-VEE WINE AND SPIRITS / BOONE	1111 8TH ST	BOONE	50036	08	BOONE	1031000	VODKA 80 PROOF	260	DIAGEO AMERICAS	37426	POPOV VODKA 80 PI		
524944000011	2015-04-08T00:00:00.000	4674	KUM & GO #227 / AMES	2108 ISSAC NEWTON DR	AMES	50010	85	STORY	1011300	TENNESSEE WHISKIES	85	BROWN-FORMAN CORPORATION	26821	JACK DANIELS OLD		
505429500008	2012-05-08T00:00:00.000	2657	HY-VEE FOOD STORE / BEDFORD	1604 BENT	BEDFORD	50033	87	TAYLOR	1011300	TENNESSEE WHISKIES	85	BROWN-FORMAN CORPORATION	26586	GENTLEMAN JACK		
520927400119	2014-08-20T00:00:00.000	3456	QUICK SHOP / CLEAR LAKE	904 W 8TH ST	CLEAR LAKE	50428	17	CERRO GORDO	1012100	CANADIAN WHISKIES	434	LUXCO-ST LOUIS	13037	CANADIAN RESERVE		
511007100026	2013-03-11T00:00:00.000	4265	KWIK STOP 3 / WATERLOO	1104 WASHINGTON ST	WATERLOO	50702	07	BLACK HAWK	1031200	VODKA FLAVORED	380	PHILLIPS BEVERAGE COMPANY	40304	UV CANDY BAR		
505932400045	2012-06-07T00:00:00.000	4612	ATC LIQUOR	214, MYRTLE ST	DAVENPORT	52002	82	SCOTT	1012100	CANADIAN WHISKIES	260	DIAGEO AMERICAS	11290	CROWN ROYAL CANA		
516910500195	2014-01-20T00:00:00.000	2506	HY-VEE #1844 / BURLINGTON	3140 AGENCY	BURLINGTON	52601	29	DES MOINES	1032200	IMPORTED VODKA - MISC	370	PERNOD RICARD USA/AUSTIN NICHOLS	34061	ABSOLUT RUBY RED		
516348400025	2013-12-18T00:00:00.000	3855	DYMO'S WINE AND SPIRITS / STORM LAKE	1201 1/2 LAKESHORE DR	STORM LAKE	50588	11	BUENA VISTA	1041100	AMERICAN DRY GINS	434	LUXCO-ST LOUIS	31654	PARAMOUNT GIN		
515261800031	2013-10-22T00:00:00.000	4545	FILL R UP	14151 WABASH AVE	COUNCIL BLUFFS	51503	78	POTTAWATTAMIE	1011200	STRAIGHT BOURBON WHISKIES	65	JIM BEAM BRANDS	19063	JIM BEAM		
517432700012	2014-02-17T00:00:00.000	4839	DS SERVICES LLC	87 8TH ST N	NORTHWOOD	50459	98	WORTH	1031000	VODKA 80 PROOF	434	LUXCO-ST LOUIS	36308	HAWKEYE VODKA		
520970000060	2014-09-02T00:00:00.000	2353	CRESCO LIQUOR STORE	700 2ND AVE SE	CRESCO	52136	45	HOWARD	1031200	VODKA FLAVORED	380	PHILLIPS BEVERAGE COMPANY	41707	PHILLIPS LINE FL		
506804700001	2012-07-26T00:00:00.000	2588	HY-VEE FOOD AND DRUG #6 / CEDAR RAPIDS	4035 MT. VERNON ROAD S.E.	CEDAR RAPIDS	52403	57	LINN	1002300	FLAVORED RUM	370	PERNOD RICARD USA/AUSTIN NICHOLS	42716	MALIBU COCONUT RI		
509346100073	2012-12-06T00:00:00.000	2637	HY-VEE #5 / DAVENPORT	2351 W LOCUST	DAVENPORT	52004	82	SCOTT	1011100	BLENDED WHISKIES	260	DIAGEO AMERICAS	25606	SEAGRAMS 7 CROWN		
509830700021	2013-01-02T00:00:00.000	3762	WINE AND SPIRITS GALLERY	7690 HICKMAN RD	WINDSOR HEIGHTS	50322	77	POLK	1012200	SCOTCH WHISKIES	55	SAZERAC NORTH AMERICA	8824	LAUDER'S		
521205800014	2014-09-15T00:00:00.000	4641	KUM & GO #573 / SE 14TH DM	5830 SE 14TH ST	DES MOINES	50315	77	POLK	1011200	STRAIGHT BOURBON WHISKIES	65	JIM BEAM BRANDS	19476	MAKER'S MARK		
519622300077	2014-06-18T00:00:00.000	4129	CYCLONE LIQUORS	626 LINCOLN WAY	AMES	50010	85	STORY	1032200	IMPORTED VODKA - MISC	370	PERNOD RICARD USA/AUSTIN NICHOLS	34029	ABSOLUT CITRON (I		
507078700019	2012-08-13T00:00:00.000	4637	KWIK SHOP #560 / CEDAR RAPIDS	3301 J ST	CEDAR RAPIDS	52404	57	LINN	1031200	VODKA FLAVORED	380	PHILLIPS BEVERAGE COMPANY	41601	UV BLUE (RASPBRI		
511231900029	2013-03-21T00:00:00.000	4345	LUCKY MART	1031 WEST ST	GRINWELL	50112	79	POMESHIEK	1031000	VODKA 80 PROOF	380	PHILLIPS BEVERAGE COMPANY	37348	PHILLIPS VODKA		
506669700004	2012-07-23T00:00:00.000	3352	CORK 'N' BOTTLE	900 EAST MAIN ST.	MANCHESTER	52057	28	DELAWARE	1011200	STRAIGHT BOURBON WHISKIES	461	CAMPARI(SKYY)	22156	WILD TURKEY 101		
513352200052	2013-07-15T00:00:00.000	2539	HY-VEE FOOD STORE / IOWA FALLS	HIGHWAY 65 SOUTH	IOWA FALLS	50126	42	HARDIN	1002200	PUERTO RICO & VIRGIN ISLANDS RUM	35	BACARDI U.S.A., INC.	43046	BACARDI SELECT RI		
512050000005	2013-05-07T00:00:00.000	4867	SPEEDY GAS N SHOP	430, S 35TH ST	COUNCIL BLUFFS	51501	78	POTTAWATTAMIE	1051010	AMERICAN GRAPE BRANDIES	115	CONSTELLATION WINE COMPANY, INC.	53211	PAUL HASSON VS BI		
510340000145	2013-02-05T00:00:00.000	3809	BOOTLEGGIN' BARZINI'S FIN	412 1ST AVE	CORALVILLE	52241	52	JOHNSON	1041100	AMERICAN DRY GINS	370	PERNOD RICARD USA/AUSTIN NICHOLS	32237	SEAGRAMS EXTRA DI		

```
CREATE TABLE Staging_Sales (  
    invoice_line_no TEXT,  
    date TEXT,  
    store TEXT,  
    name TEXT,  
    address TEXT,  
    city TEXT,  
    zipcode TEXT,  
    store_location TEXT,  
    county_number TEXT,  
    county TEXT,  
    category TEXT,  
    category_name TEXT,  
    vendor_no TEXT,  
    vendor_name TEXT,  
    itemno TEXT,  
    im_desc TEXT,  
    pack TEXT,  
    bottle_volume_ml TEXT,  
    state_bottle_cost TEXT,  
    state_bottle_retail TEXT,  
    sale_bottles TEXT,  
    sale_dollars TEXT,  
    sale_liters TEXT,  
    sale_gallons TEXT,  
    file_name TEXT NOT NULL,  
    processed_timestamp DATETIME NOT NULL,  
    record_source TEXT NOT NULL  
);
```

Data Transformation

Data Transformation

❖ Case Study: Iowa Liquor Sale



CDC



Change Data
Capture

```
sale_gallons TEXT,  
file_name TEXT NOT NULL,  
processed_timestamp DATETIME NOT NULL,  
record_source TEXT NOT NULL  
);
```

Data
Cleaning



Duplicated

Missing

Type
Casting

Filtering

Invalid

Data
Processing

Enrichment

Aggregate

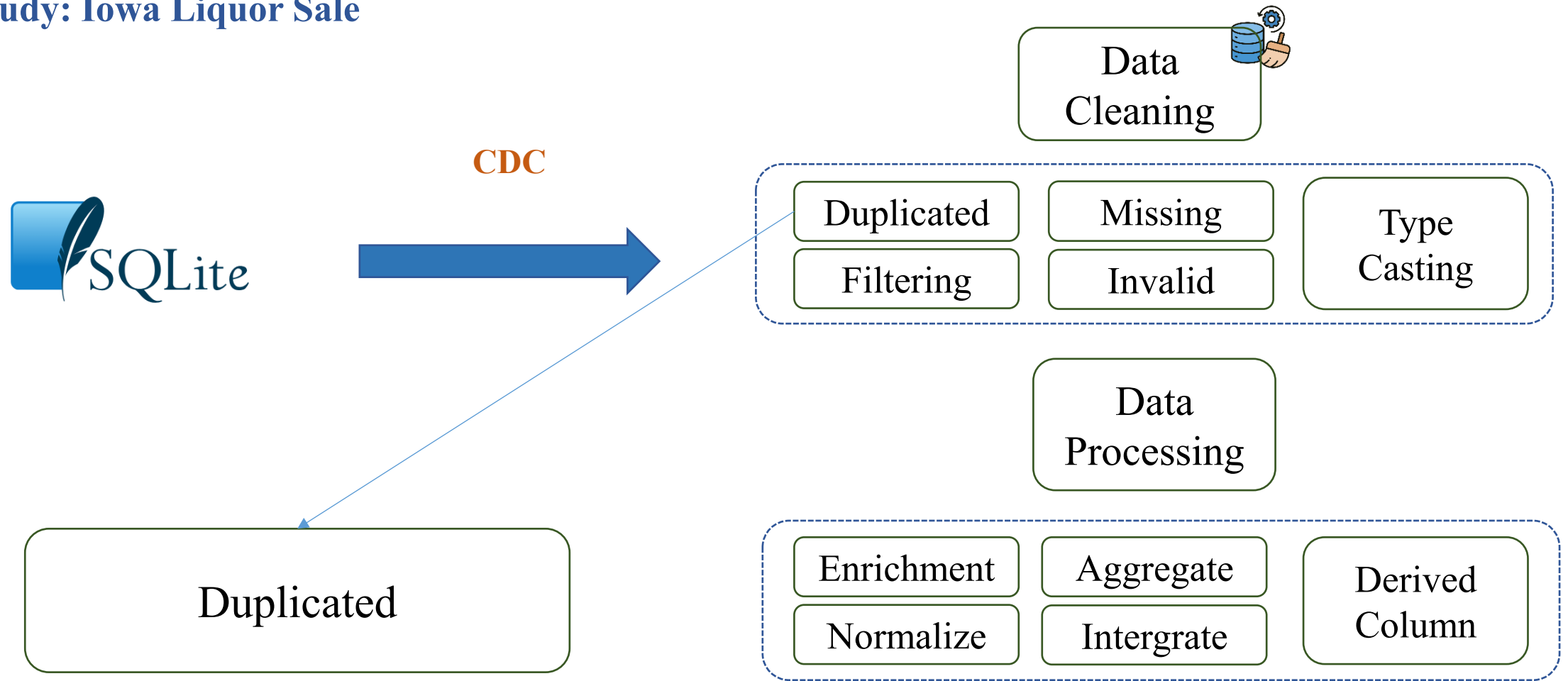
Derived
Column

Normalize

Intergrate

Data Transformation

❖ Case Study: Iowa Liquor Sale





❖ Ful Duplicated Samples

```
df_full_duplicated = df[df.duplicated(keep=False)]  
print(len(df_full_duplicated))  
print(len(df_full_duplicated.drop_duplicates()))  
df_full_duplicated.head()
```



2018
1009

	invoice_line_no	date	store	name	address	city	zipcode	store_location	county_number	county	...	itemno	im_desc	pac
14000	INV-15181500011	2018-10-22T00:00:00.000	5364	STATION MART	3594 LAFAYETTE ROAD	EVANSDALE	50707.0	{'type': 'Point', 'coordinates': [-92.287994, ...]}	07	BLACK HAWK	...	86668	JACK DANIELS TENNESSEE HONEY MINI	12.
14001	INV-15172700084	2018-10-22T00:00:00.000	2578	HY-VEE / CHARLES CITY	901 KELLY ST	CHARLES CITY	50616.0	{'type': 'Point', 'coordinates': [-92.675068, ...]}	34	FLOYD	...	43121	BACARDI SUPERIOR MINI	12.
14002	INV-15112800194	2018-10-17T00:00:00.000	2625	HY-VEE WINE & SPIRITS #2 / DAVENPORT	3301 W KIMBERLY RD	DAVENPORT	52804.0	NaN	82	SCOTT	...	69658	DR. MCGILLICUDDY'S MENTHOLMINT	6.
14003	INV-15217800109	2018-10-23T00:00:00.000	4167	IOWA STREET MARKET, INC.	1256 IOWA ST	DUBUQUE	52001.0	{'type': 'Point', 'coordinates': [-90.668138, ...]}	31	DUBUQUE	...	34505	SOBIESKI	12.
14004	INV-15176500004	2018-10-22T00:00:00.000	4051	FAREWAY STORES #639 / MAQUOKETA	110 WESTGATE DR	MAQUOKETA	52060.0	{'type': 'Point', 'coordinates': [-90.680623, ...]}	49	JACKSON	...	73055	RUMCHATA	12.

5 rows × 24 columns



Data Transformation

❖ Special Duplicated Sample

```
df_duplicated = df[df.duplicated(subset=['invoice_line_no', 'store'], keep=False)]  
print(len(df_duplicated))  
df_duplicated.tail()
```



5872

	invoice_line_no	date	store	name	address	city	zipcode	store_location	county_number	county	...	itemno	im_desc	pack	bc
99995	INV-15612400045	2018-11-09T00:00:00.000	4183	FAREWAY STORES #589 / DECORAH	103 E MAIN	DECORAH	52101.0	{'type': 'Point', 'coordinates': [-91.785286, ...]}	96.0	WINNESHIEK	...	82847	DEKUYPER LUSCIOUS PEACHTREE	12.0	
99996	INV-15672400041	2018-11-13T00:00:00.000	2624	HY-VEE #2 / DUBUQUE	2395 NW ARTERIAL RD	DUBUQUE	52002.0	{'type': 'Point', 'coordinates': [-90.741064, ...]}	31.0	DUBUQUE	...	39669	360 LIME MINI	6.0	
99997	INV-15611700005	2018-11-09T00:00:00.000	5537	PRONTO MARKET / SUMNER	117 W 1ST ST	SUMNER	50674.0	{'type': 'Point', 'coordinates': [-92.098713, ...]}	9.0	BREMER	...	11346	SEAGRAMS VO CANADIAN WHISKEY	12.0	
99998	INV-15642300070	2018-11-12T00:00:00.000	3562	WAL-MART 0797 / W BURLINGTON	324 WEST AGENCY RD	WEST BURLINGTON	52655.0	{'type': 'Point', 'coordinates': [-91.158407, ...]}	29.0	DES MOINES	...	63812	TGIFRIDAYS ULTIMATE MUDSLIDE	6.0	
99999	INV-15746000092	2018-11-15T00:00:00.000	2595	HY-VEE WINE AND SPIRITS / DENISON	1620 4TH AVE, SOUTH	DENISON	51442.0	{'type': 'Point', 'coordinates': [-95.348601, ...]}	24.0	CRAWFORD	...	56196	PAUL MASSON PEACH GRANDE AMBER BRANDY	12.0	



❖ Duplicated Sample



```
# Check example of duplicattion
```

```
df_duplicated[(df_duplicated["store"] == 5364) & (df_duplicated["invoice_line_no"] == "INV-15181500011")]
```



	invoice_line_no	date	store	name	address	city	zipcode	store_location	county_number	county	...	itemno	im_desc	pack	bottle_volume_ml	st
14000	INV-15181500011	2018-10-22T00:00:00.000	5364	STATION MART	3594 LAFAYETTE ROAD	EVANSDALE	50707.0	{'type': 'Point', 'coordinates': [-92.287994, ...]}	07	BLACK HAWK	...	86668	JACK DANIELS TENNESSEE HONEY MINI	12.0	500.0	
22820	INV-15181500011	2018-10-22T00:00:00.000	5364	STATION MART	3594 LAFAYETTE ROAD	EVANSDALE	50707.0	{'type': 'Point', 'coordinates': [-92.287994, ...]}	07	BLACK HAWK	...	86668	JACK DANIELS TENNESSEE HONEY MINI	12.0	500.0	

2 rows × 24 columns



```
df_duplicated.drona( )
```



❖ Updated Sample

```
df_duplicated[(df_duplicated["store"] == 4183) & (df_duplicated["invoice_line_no"] == "INV-15612400045")]
```



	invoice_line_no	date	store	name	address	city	zipcode	store_location	county_number	county	...	itemno	im_desc	pack	bottle_volume_ml	state_bottle_cost
75123	INV-15612400045	2018-11-09T00:00:00.000	4183	FAREWAY STORES #589 / DECORAH	103 E MAIN	DECORAH	52101.0	96	WINNESHIEK	1081400	...	DEKUYPER LUSCIOUS PEACHTREE	12	1000.0	7.87	11.81
99995	INV-15612400045	2018-11-09T00:00:00.000	4183	FAREWAY STORES #589 / DECORAH	103 E MAIN	DECORAH	52101.0	{'type': 'Point', 'coordinates': [-91.785286, ...]}	96.0	WINNESHIEK	...	82847	DEKUYPER LUSCIOUS PEACHTREE	12.0	1000.00	7.87

2 rows × 24 columns

```
df_duplicated[(df_duplicated["store"] == 3562) & (df_duplicated["invoice_line_no"] == "INV-15642300070")]
```



	invoice_line_no	date	store	name	address	city	zipcode	store_location	county_number	county	...	itemno	im_desc	pack	bottle_volume_ml	state_bottle_cost
75126	INV-15642300070	2018-11-12T00:00:00.000	3562	WAL-MART 0797 / W BURLINGTON	324 WEST AGENCY RD	WEST BURLINGTON	52655.0	29	DES MOINES	1071100	...	TGIFRIDAYS ULTIMATE MUDSLIDE	6	1750.0	7.52	11.28
99998	INV-15642300070	2018-11-12T00:00:00.000	3562	WAL-MART 0797 / W BURLINGTON	324 WEST AGENCY RD	WEST BURLINGTON	52655.0	{'type': 'Point', 'coordinates': [-91.158407, ...]}	29.0	DES MOINES	...	63812	TGIFRIDAYS ULTIMATE MUDSLIDE	6.0	1750.00	7.52

2 rows × 24 columns



Data Transformation

❖ Updated Sample

```
df_update = df[df.duplicated(subset=['invoice_line_no', 'store'])]
```

```
df_duplicated[(df_duplicated["store"] == 3562) & (df_duplicated["invoice_line_no"] == "INV-15642300070")]
```



	invoice_line_no	date	store	name	address	city	zipcode	store_location	county_number	county	...	itemno	im_desc	pack	bottle_volume_ml	state_bottle_cost
75126	INV-15642300070	2018-11-12T00:00:00.000	3562	WAL-MART 0797 / W BURLINGTON	324 WEST AGENCY RD	WEST BURLINGTON	52655.0	29	DES MOINES	1071100	...	TGIFRIDAYS ULTIMATE MUDSLIDE	6	1750.0	7.52	11.28
99998	INV-15642300070	2018-11-12T00:00:00.000	3562	WAL-MART 0797 / W BURLINGTON	324 WEST AGENCY RD	WEST BURLINGTON	52655.0	{'type': 'Point', 'coordinates': [-91.158407, ...]}	29.0	DES MOINES	...	63812	TGIFRIDAYS ULTIMATE MUDSLIDE	6.0	1750.00	7.52

2 rows × 24 columns

```
df_update[df_update['invoice_line_no'] == "INV-15642300070"]
```

	invoice_line_no	date	store	name	address	city	zipcode	store_location	county_number	county	...	itemno	im_desc	pack	bottle
99998	INV-15642300070	2018-11-12T00:00:00.000	3562	WAL-MART 0797 / W BURLINGTON	324 WEST AGENCY RD	WEST BURLINGTON	52655.0	{'type': 'Point', 'coordinates': [-91.158407, ...]}	29.0	DES MOINES	...	63812	TGIFRIDAYS ULTIMATE MUDSLIDE	6.0	

1 rows × 24 columns



❖ Drop Duplicated Samples and Keep Updated Samples

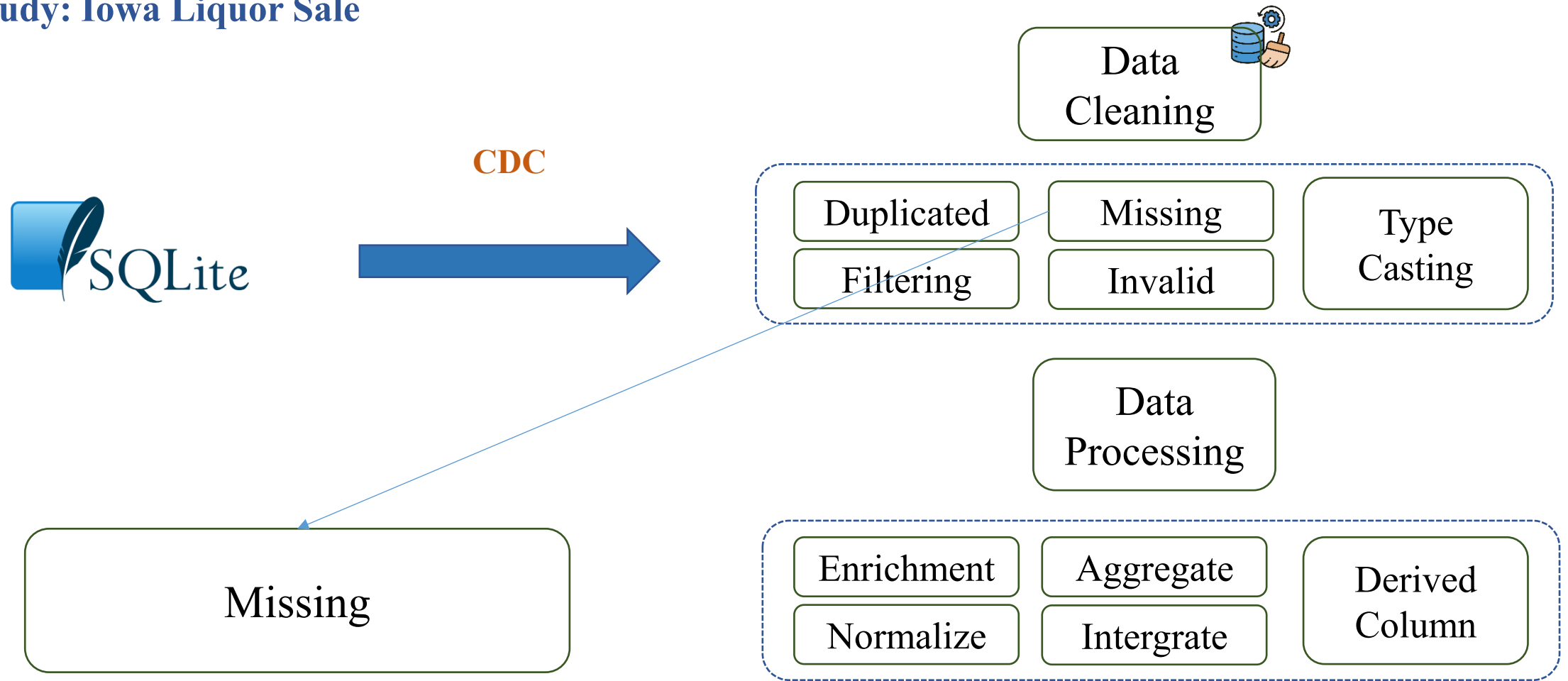
```
df = df[df.duplicated(subset=['invoice_line_no', 'store'], keep='first') | ~df.duplicated(subset=['invoice_line_no', 'store'], keep=False)]
```

```
df[df['invoice_line_no'] == "INV-15642300070"]
```

	invoice_line_no	date	store	name	address	city	zipcode	store_location	county_number	county	...	itemno	im_desc	pack	b
99998	INV-15642300070	2018-11-12T00:00:00.000	3562	WAL-MART 0797 / W BURLINGTON	324 WEST AGENCY RD	WEST BURLINGTON	52655.0	{'type': 'Point', 'coordinates': [-91.158407, ...	29.0	DES MOINES	...	63812	TGIFRIDAYS ULTIMATE MUDSLIDE	6.0	

Data Transformation

❖ Case Study: Iowa Liquor Sale



❖ Handle Null Data

```
['address',  
 'city',  
 'zipcode',  
 'store_location',  
 'county_number',  
 'county',  
 'category',  
 'category_name',  
 'state_bottle_cost',  
 'state_bottle_retail',  
 'sale_bottles',  
 'sale_dollars',  
 'sale_liters',  
 'sale_gallons']
```

```
df_null = df[df.isnull().any(axis=1)]  
df_null.info()
```

```
null_columns = df.columns[df.isnull().sum() > 0].tolist()
```

Index: 9369 entries, 2 to 99976
Data columns (total 24 columns):

#	Column	Non-Null Count	Dtype
0	invoice_line_no	9369 non-null	object
1	date	9369 non-null	object
2	store	9369 non-null	int64
3	name	9369 non-null	object
4	address	8908 non-null	object
5	city	8908 non-null	object
6	zipcode	8908 non-null	object
7	store_location	1281 non-null	object
8	county_number	8908 non-null	object
9	county	8914 non-null	object
10	category	9191 non-null	object
11	category_name	9205 non-null	object
12	vendor_no	9369 non-null	object
13	vendor_name	9369 non-null	object
14	itemno	9369 non-null	object
15	im_desc	9369 non-null	object
16	pack	9369 non-null	float64
17	bottle_volume_ml	9369 non-null	float64
18	state_bottle_cost	9356 non-null	object
19	state_bottle_retail	9356 non-null	object
20	sale_bottles	9356 non-null	float64
21	sale_dollars	9285 non-null	object
22	sale_liters	9356 non-null	float64
23	sale_gallons	8306 non-null	object

dtypes: float64(4), int64(1), object(19)
memory usage: 1.8+ MB



Data Transformation

❖ Full Duplicated Samples

['address',
'city',
'zipcode',
'store_location',
'county_number',
'county',
'category',
'category name',
'state_bottle_cost',
'state_bottle_retail',
'sale_bottles',
'sale_dollars',
'sale_liters',
'sale_gallons']

Fill by "Unknown"

```
df.fillna(value={'address': 'Unknown',  
                'city': 'Unknown',  
                'zipcode': 'Unknown',  
                'county_number': 'Unknown',  
                'county': 'Unknown',  
                'category': 'Unknown',  
                'category_name': 'Unknown'}, inplace=True)
```

**Drop Null
Values**

```
df = df.dropna(subset=['state_bottle_cost', 'state_bottle_retail', 'sale_bottles', 'sale_dollars', 'sale_liters',  
                      'sale_gallons'])
```

Index: 9369 entries, 2 to 99976

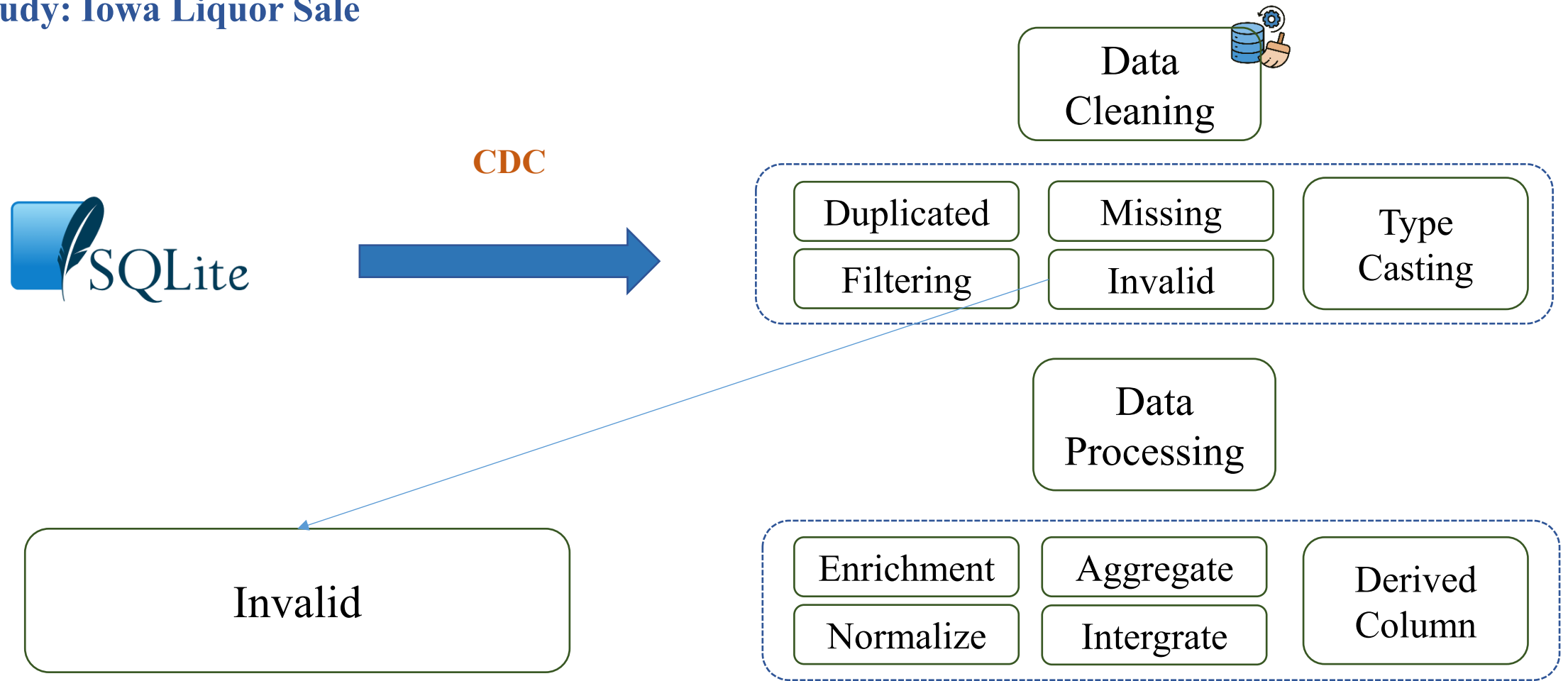
Data columns (total 24 columns):

#	Column	Non-Null	Count	Dtype
0	invoice_line_no	9369	non-null	object
1	date	9369	non-null	object
2	store	9369	non-null	int64
3	name	9369	non-null	object
4	address	8908	non-null	object
5	city	8908	non-null	object
6	zipcode	8908	non-null	object
7	store_location	1281	non-null	object
8	county_number	8908	non-null	object
9	county	8914	non-null	object
10	category	9191	non-null	object
11	category_name	9205	non-null	object
12	vendor_no	9369	non-null	object
13	vendor_name	9369	non-null	object
14	itemno	9369	non-null	object
15	im_desc	9369	non-null	object
16	pack	9369	non-null	float64
17	bottle_volume_ml	9369	non-null	float64
18	state_bottle_cost	9356	non-null	object
19	state_bottle_retail	9356	non-null	object
20	sale_bottles	9356	non-null	float64

**Drop
column**

Data Transformation

❖ Case Study: Iowa Liquor Sale





❖ Invalid by Logic

Amount Logic

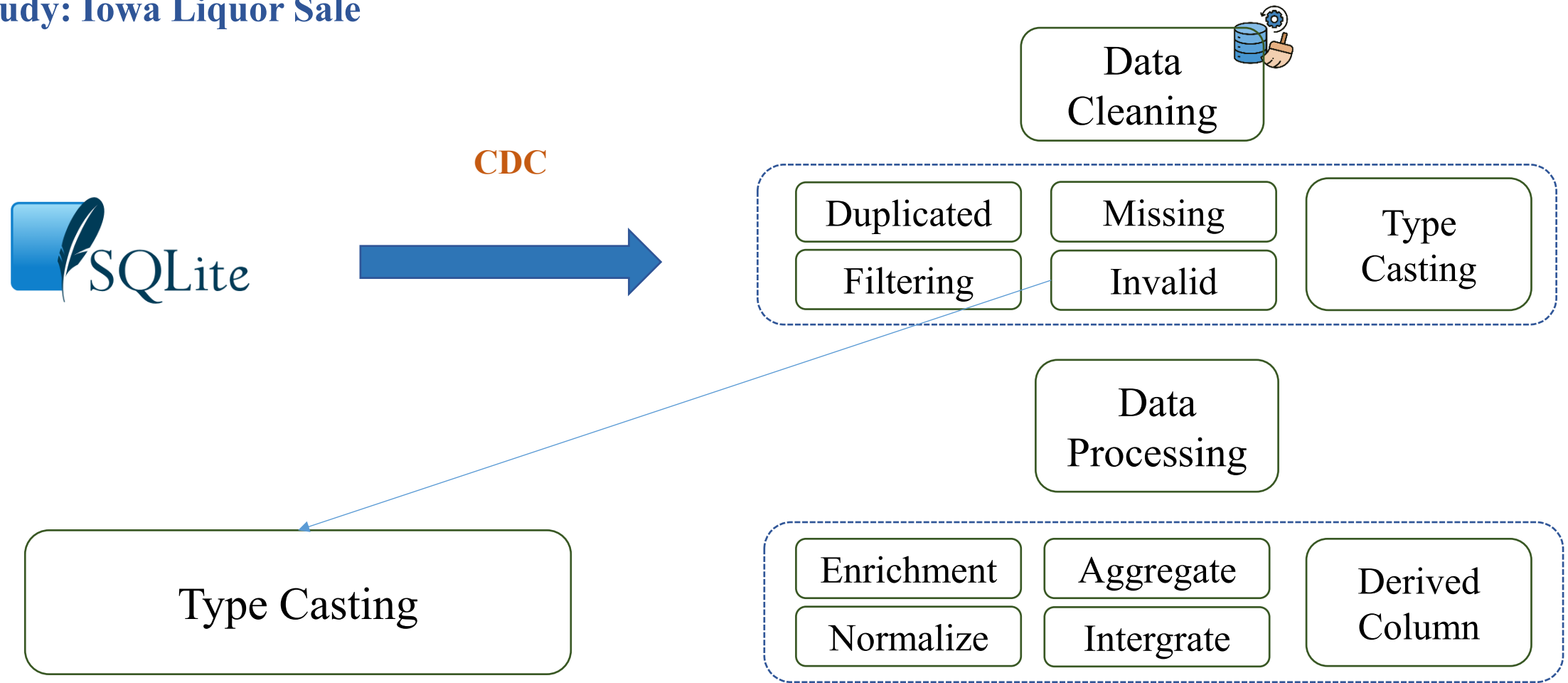
pack	bottle_volume_ml	state_bottle_cost	state_bottle_retail	sale_bottles	sale_dollars	sale_liters	sale_gallons
6.0	1750.0	10.49	22.49	12.0	269.88	12.0	3.17
12.0	1000.0	14.99	9.71	6.0	58.26	4.5	1.18
12.0	750.0	6.47	10.76	6.0	64.56	10.5	2.77
6.0	1750.0	7.17	19.65	12.0	235.80	9.0	2.37



```
numeric_cols = ['pack', 'bottle_volume_ml', 'state_bottle_cost', 'state_bottle_retail',  
                'sale_bottles', 'sale_dollars', 'sale_liters', 'sale_gallons']  
  
# Convert columns to numeric  
for col in numeric_cols:  
    df[col] = pd.to_numeric(df[col])  
  
df_invalid = df[(df[numeric_cols] < 1).all(axis=1)]
```

Data Transformation

❖ Case Study: Iowa Liquor Sale





Data Transformation

❖ Type Casting

```
numeric_cols = ['state_bottle_cost', 'state_bottle_retail', 'sale_bottles',  
                'sale_dollars', 'sale_liters', 'sale_gallons']  
for col in numeric_cols:  
    df[col] = pd.to_numeric(df[col], errors='coerce')
```

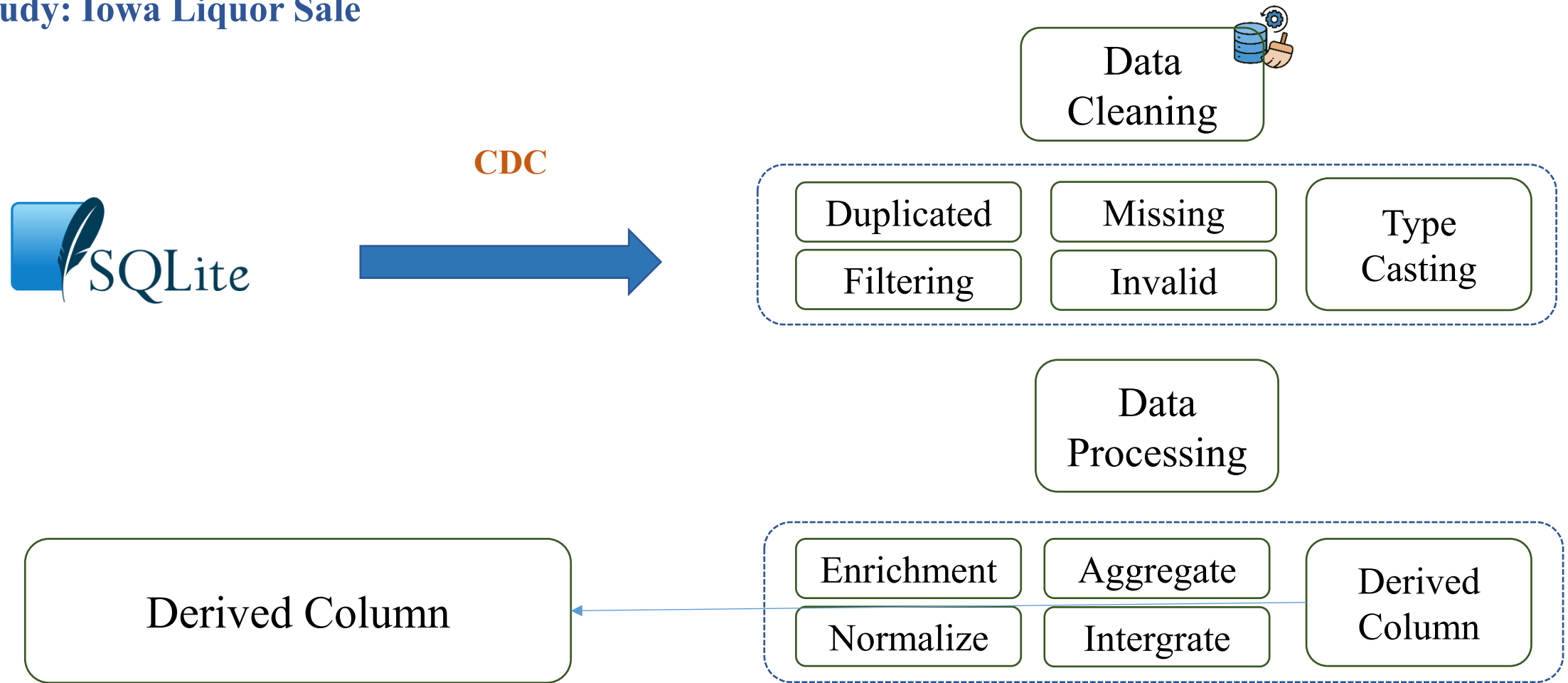
Convert to Numeric Data

```
df['date'] = pd.to_datetime(df['date'], errors='coerce')
```

Type Casting -> Valid Date Type

Data Transformation

❖ Case Study: Iowa Liquor Sale





Data Transformation

❖ Data processing: Business Logic

Derived
Column

store_id	transac_id	amount	cost
st00a	tran001	1	2
st00a	tran002	3	2.3
st00a	tran003	2	1.9



total_cost
2
6.9
3.8



❖ Data processing: Business Logic

```
df['revenue'] = df['sale_dollars']
df['cost'] = df['state_bottle_cost'] * df['sale_bottles']
df['profit'] = (df['state_bottle_retail'] - df['state_bottle_cost']) * df['sale_bottles']
df['total_bottles_sold'] = df['sale_bottles']
df['total_volume_sold_in_liters'] = df['sale_liters']
df['profit_margin'] = (df['profit'] / df['revenue'] * 100).round(2).where(df['revenue'] > 0, 0)
df['average_bottle_price'] = (df['sale_dollars'] / df['sale_bottles']).round(2).where(df['sale_bottles'] > 0, 0)
df['volume_per_bottle_sold'] = (df['sale_liters'] / df['sale_bottles']).round(2).where(df['sale_bottles'] > 0, 0)
```

```
derived_columns = ['revenue', 'cost', 'profit', 'total_bottles_sold', 'total_volume_sold_in_liters', 'profit_margin', 'average_bottle_price', 'volume_per_bottle_sold']
df_derived = df[derived_columns]
df_derived.head()
```



	revenue	cost	profit	total_bottles_sold	total_volume_sold_in_liters	profit_margin	average_bottle_price	volume_per_bottle_sold
0	94.44	62.94	31.50	6.0	10.5	33.35	15.74	1.75
1	269.88	179.88	90.00	12.0	12.0	33.35	22.49	1.00
2	58.26	38.82	19.44	6.0	4.5	33.37	9.71	0.75
3	64.56	43.02	21.54	6.0	10.5	33.36	10.76	1.75
5	235.80	157.20	78.60	12.0	9.0	33.33	19.65	0.75



Derived
Column



Data Loading

❖ Dimension and Fact Table (Problem Define)

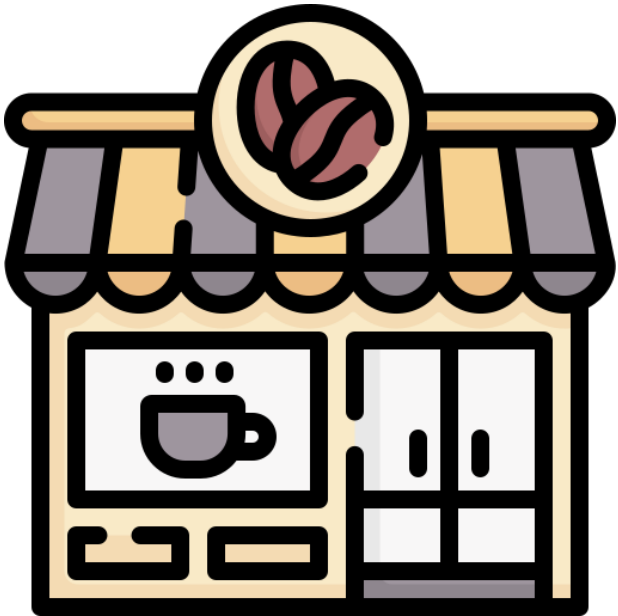


- Customer Alice buys 2 lattes for \$8 at 8 AM.
- Customer Bob buys 1 cappuccino for \$5 at 9 AM.
- Customer Alice buys 1 espresso for \$3 at 2 PM.



Data Loading

❖ Dimension and Fact Table (Problem Define)



Sale ID	Customer ID	Product ID	Date ID	Quantity	Total Amount
1	C001	P001	D001	2	\$8
2	C002	P002	D001	1	\$5
3	C001	P003	D001	1	\$3



Data Loading

❖ Dimension and Fact Table (Problem Define)

Customer ID	Name	City	Phone
C001	Alice	Seattle	555-1234
C002	Bob	Portland	555-5678

Product ID	Product Name	Category	Price
P001	Latte	Coffee	\$4
P002	Cappuccino	Coffee	\$5
P003	Espresso	Coffee	\$3

Date ID	Date	Day of Week	Month	Year
D001	2025-07-29	Tuesday	July	2025



Data Loading

❖ Dimension and Fact Table (Problem Define)

Sale ID	Customer ID	Product ID	Date ID	Quantity	Total Amount
1	C001	P001	D001	2	\$8
2	C002	P002	D001	1	\$5
3	C001	P003	D001	1	\$3

Fact tables

Fact tables store quantitative, measurable data about business events

Key Characteristics:

- Contain numerical measures
- Include foreign keys



Data Loading

❖ Dimension and Fact Table (Problem Define)

Date ID	Date	Day of Week	Month	Year
D001	2025-07-29	Tuesday	July	2025

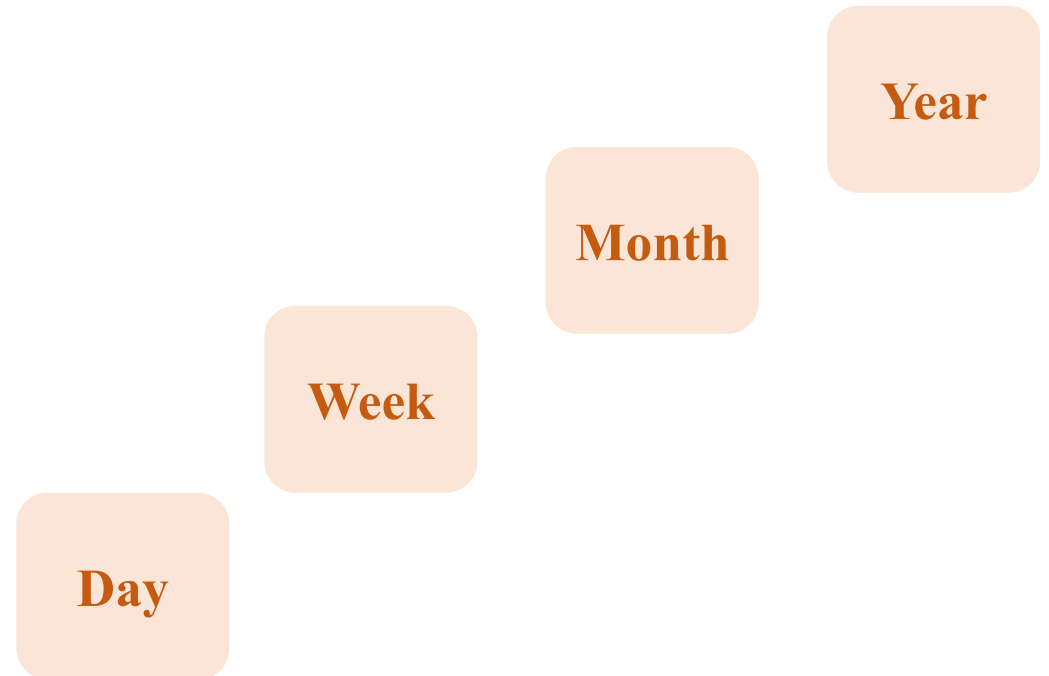
Dims tables

Hierarchy

Dimension tables store descriptive attributes that provide context to the facts, answering “who, what, where, when” questions.

Key Characteristics:

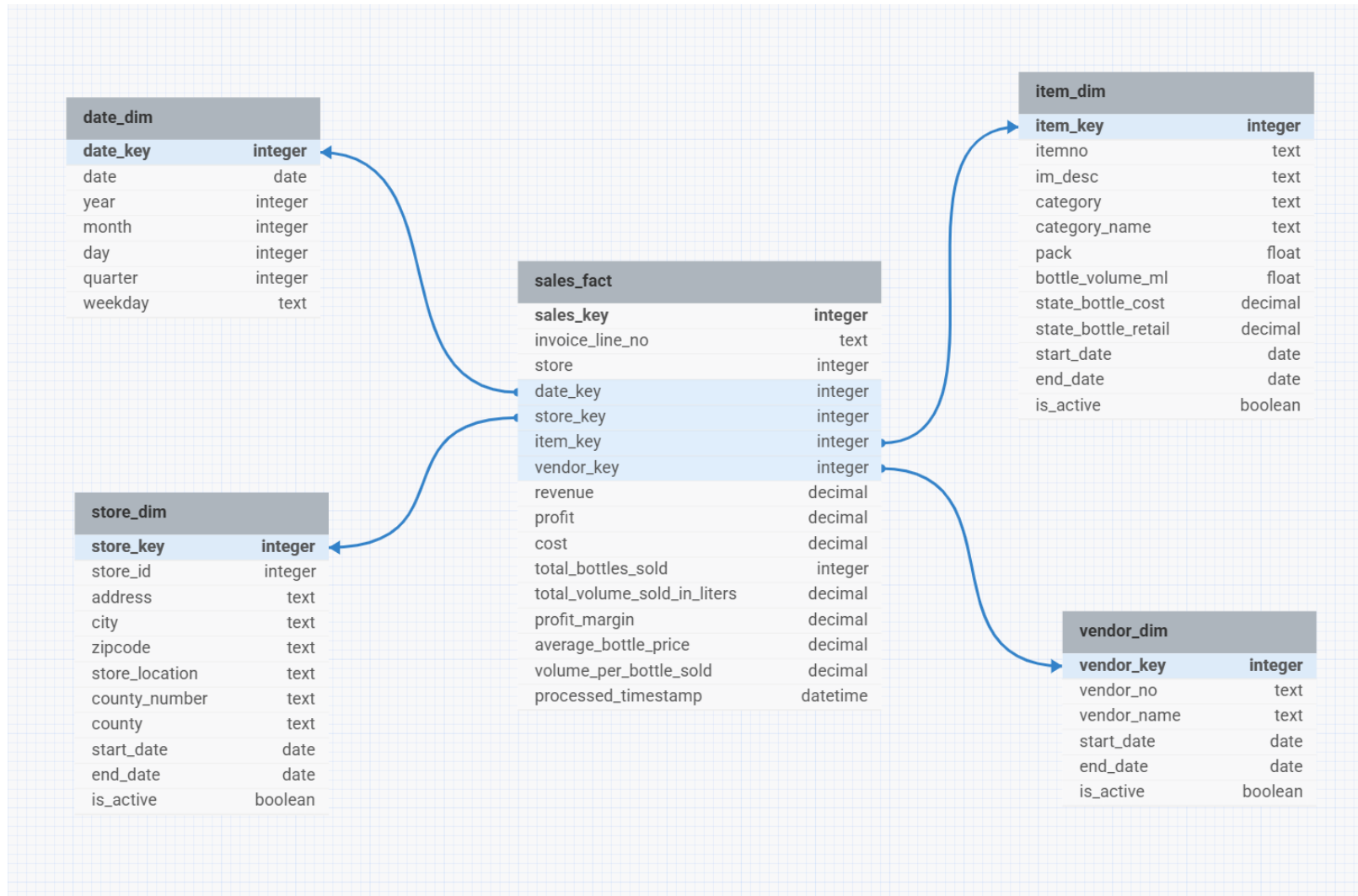
- Contain descriptive columns
- Have a primary key
- have fewer rows than fact tables





Data Loading

❖ Our Dimension and Fact Table





Data Loading

❖ Slowly Changing Dimension: Type 1 - Overwrite

Existing Record

Supplier_Key	Supplier_Code	Supplier_Name	Supplier_State
123	ABC	Acme Supply Co	CA



Update

Supplier_Key	Supplier_Code	Supplier_Name	Supplier_State
123	ABC	Acme Supply Co	IL

Updating Record



Data Loading

❖ Slowly Changing Dimension: Type 2 – Add New Row

Existing Record

Supplier_Key	Supplier_Code	Supplier_Name	Supplier_State
123	ABC	Acme Supply Co	CA



Update

Supplier_Key	Supplier_Code	Supplier_Name	Supplier_State	Version
123	ABC	Acme Supply Co	CA	0
124	ABC	Acme Supply Co	IL	1

Updating Record



Data Loading

❖ Slowly Changing Dimension: Type 2 – Add New Row

Existing Record

Supplier_Key	Supplier_Code	Supplier_Name	Supplier_State
123	ABC	Acme Supply Co	CA



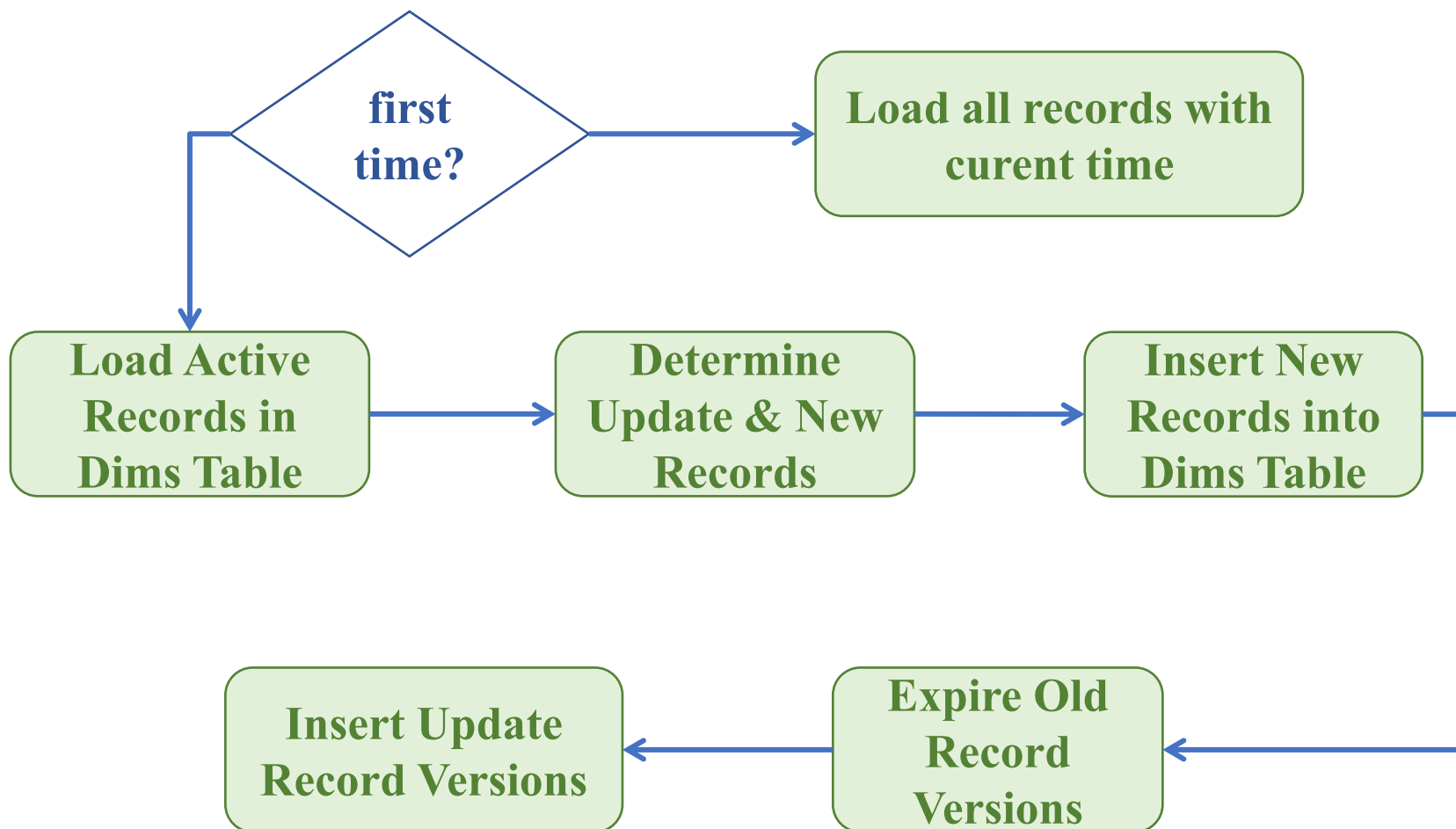
Update

Supplier_Key	Supplier_Code	Supplier_Name	Supplier_State	Start_Date	End_Date
123	ABC	Acme Supply Co	CA	2000-01-01T00:00:00	2004-12-22T00:00:00
124	ABC	Acme Supply Co	IL	2004-12-22T00:00:00	NULL

Updating Record

Data Loading

❖ Slowly Changing Dimension: Type 2 – Add New Row



Tracking Columns

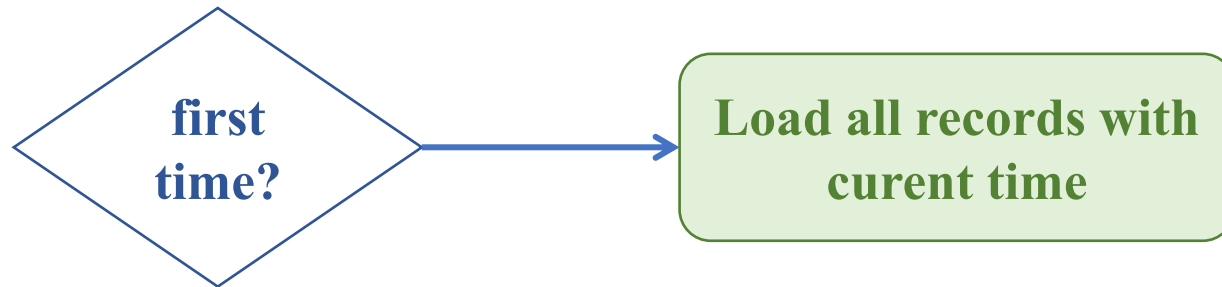
start_date

end_date

is_active = 0/1 (False/ True)

store_dim	
store_key	integer
store_id	integer
address	text
city	text
zipcode	text
store_location	text
county_number	text
county	text
start_date	date
end_date	date
is_active	boolean

❖ Slowly Changing Dimension: Type 2 – Add New Row



```
def process_scd_type2(df, dim_table, key_col, attributes, conn):  
    """Handle Slowly Changing Dimension Type 2 changes"""  
  
    # Create a copy to avoid SettingWithCopyWarning  
    df = df.copy()  
  
    if current_dim.empty:  
        # Initial load - all records are new  
        df = df.assign(  
            start_date=datetime.now().date(),  
            end_date=None,  
            is_active=True  
        )  
        df.to_sql(dim_table, conn, if_exists='append', index=False)  
    return
```

Tracking Columns

start_date

end_date

is_active = 0/1 (False/ True)

store_dim	
store_key	integer
store_id	integer
address	text
city	text
zipcode	text
store_location	text
county_number	text
county	text
start_date	date
end_date	date
is_active	boolean

❖ Slowly Changing Dimension: Type 2 – Add New Row

first
time?

Load Active
Records in
Dims Table

```
if current_dim.empty:  
    # Initial load - all records are new  
    df = df.assign(  
        start_date=datetime.now().date(),  
        end_date=None,  
        is_active=True  
    )  
    df.to_sql(dim_table, conn, if_exists='append', index=False)  
    return
```

Tracking Columns

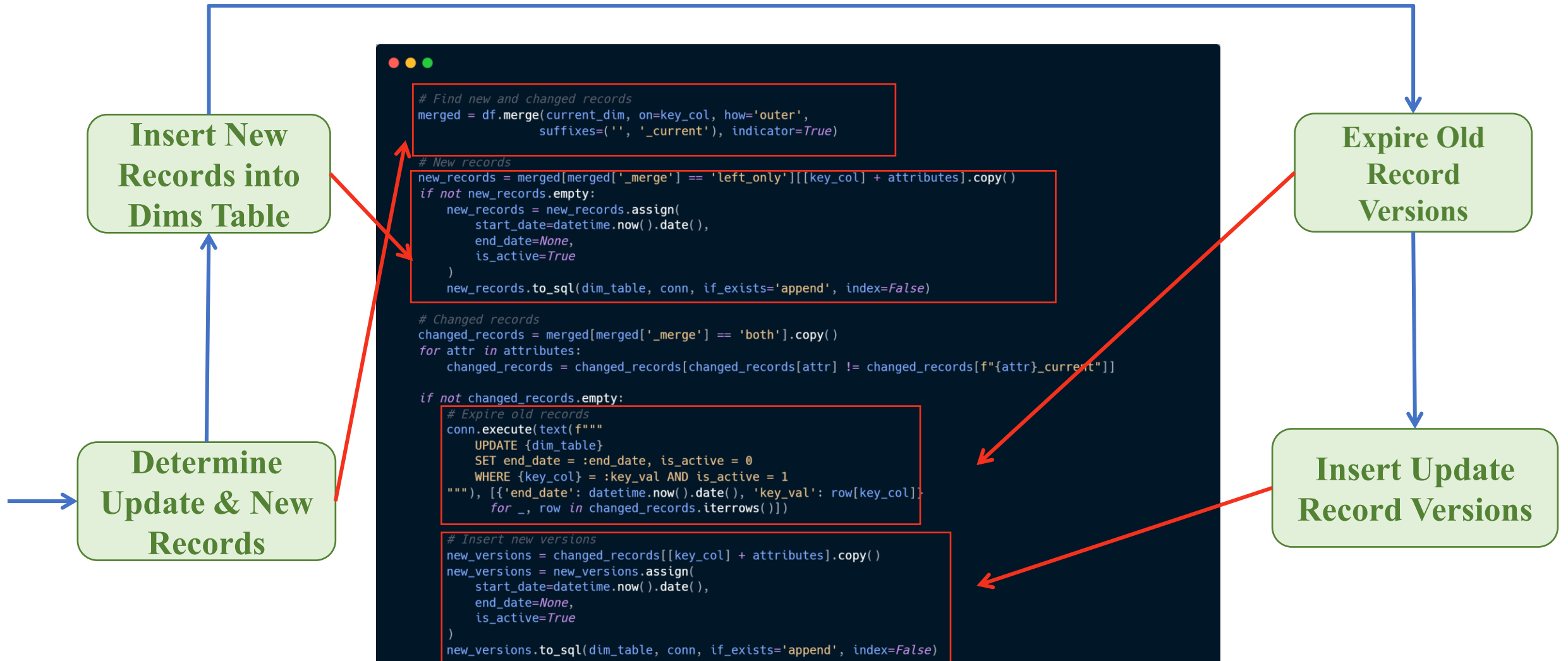
start_date

end_date

is_active = 0/1 (False/ True)

store_dim	
store_key	integer
store_id	integer
address	text
city	text
zipcode	text
store_location	text
county_number	text
county	text
start_date	date
end_date	date
is_active	boolean

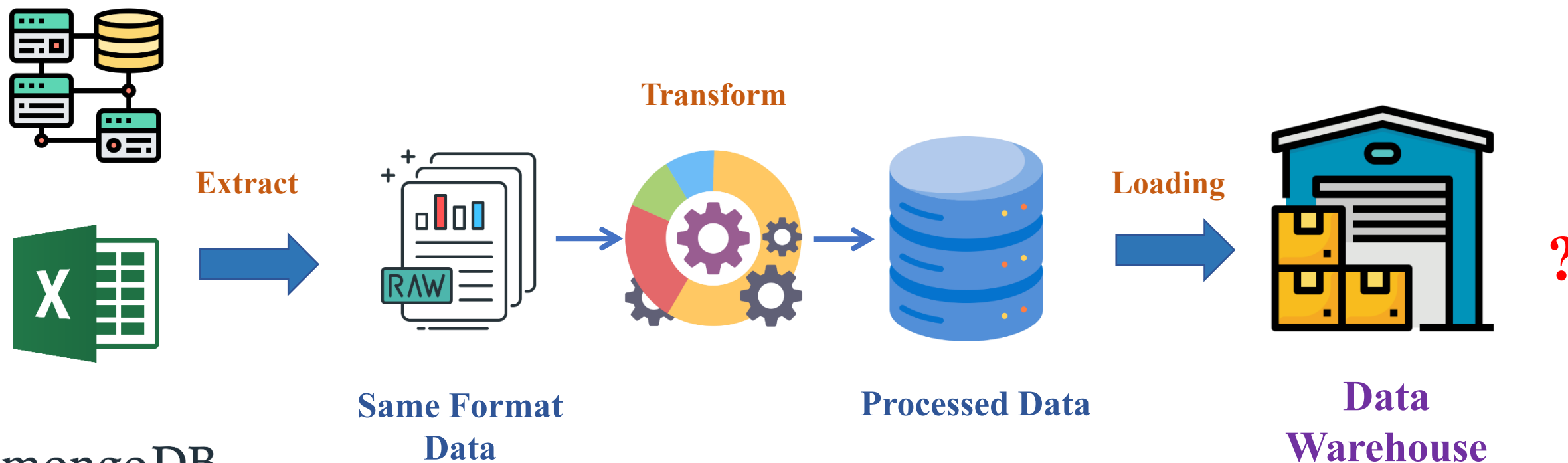
❖ Slowly Changing Dimension: Type 2 – Add New Row





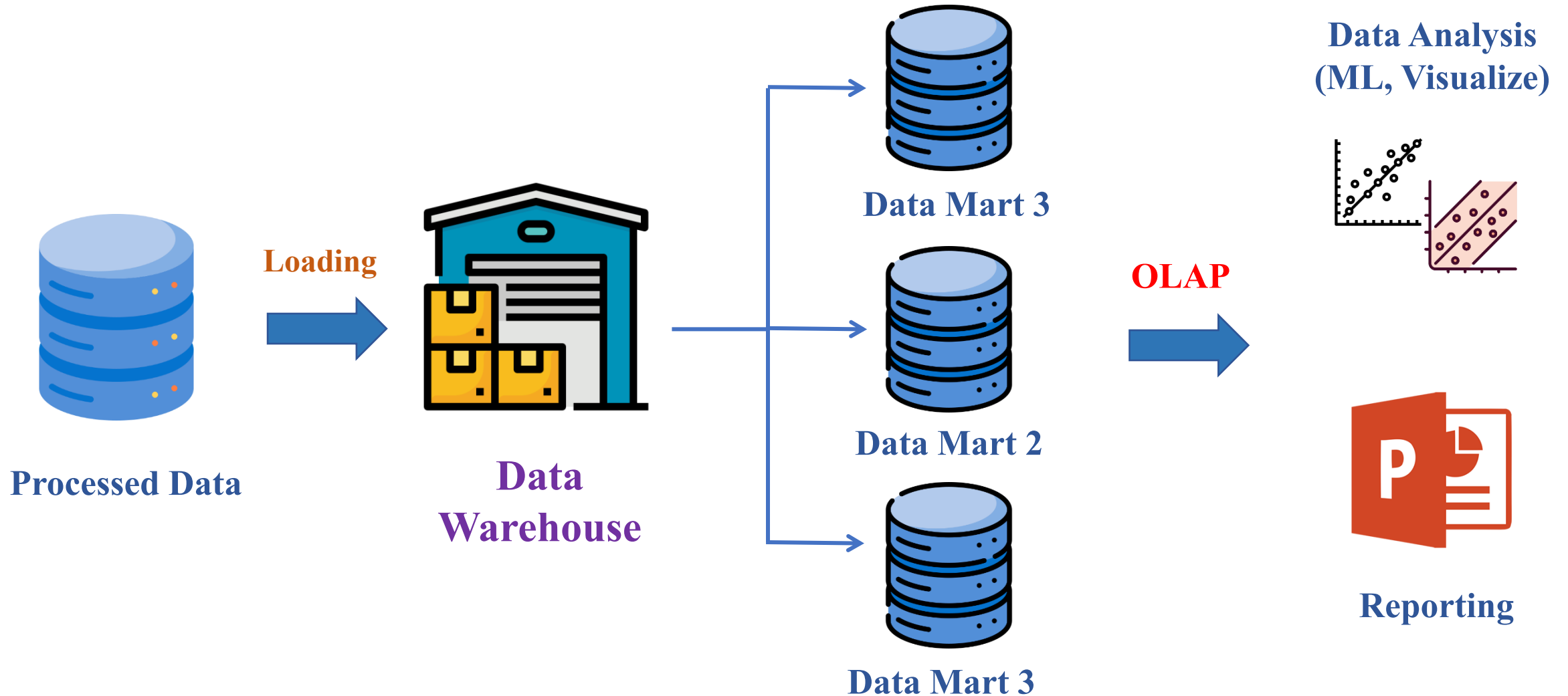
Data Loading

❖ What's next?



Data Loading

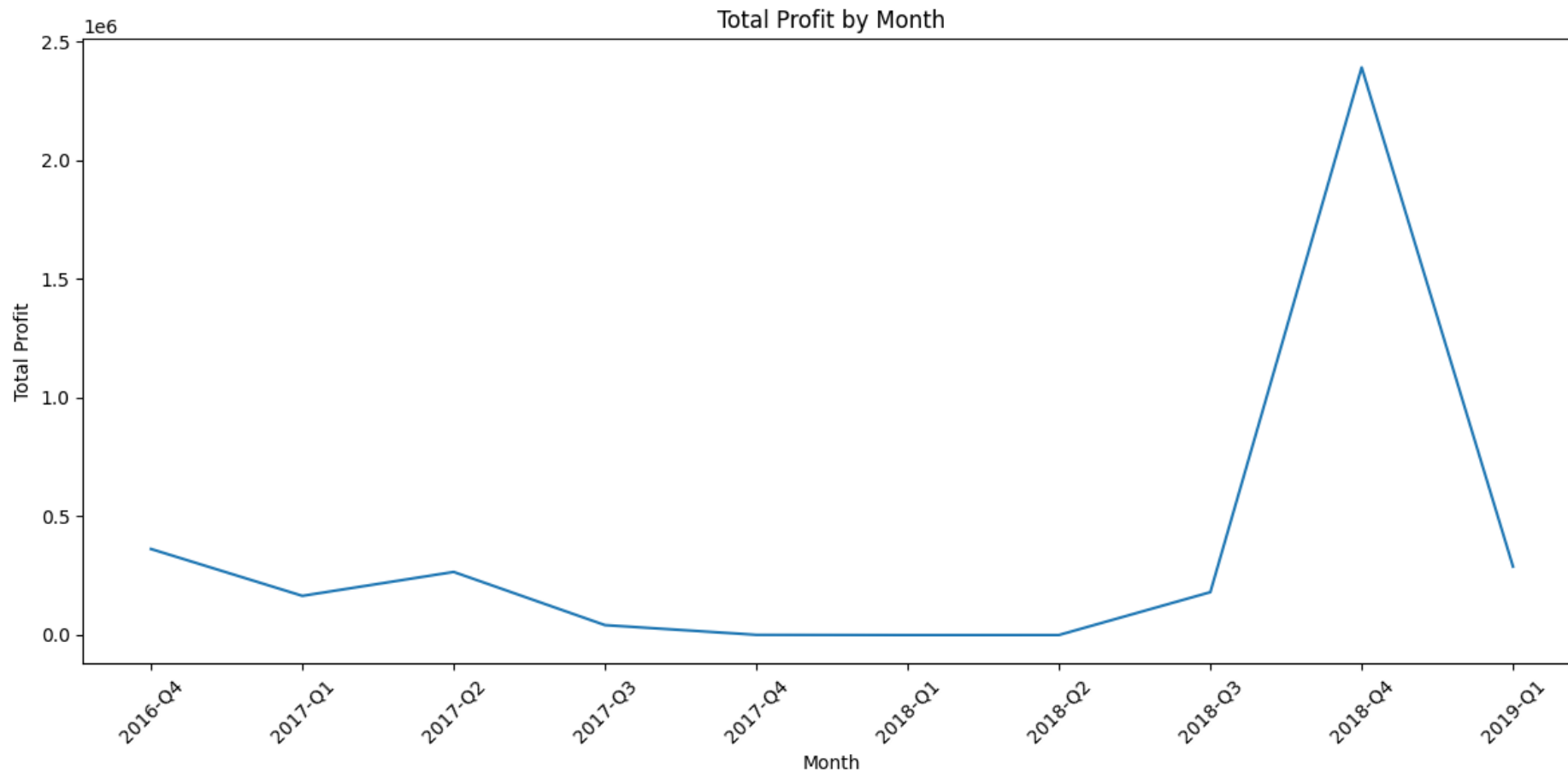
❖ What's next?





Data Loading

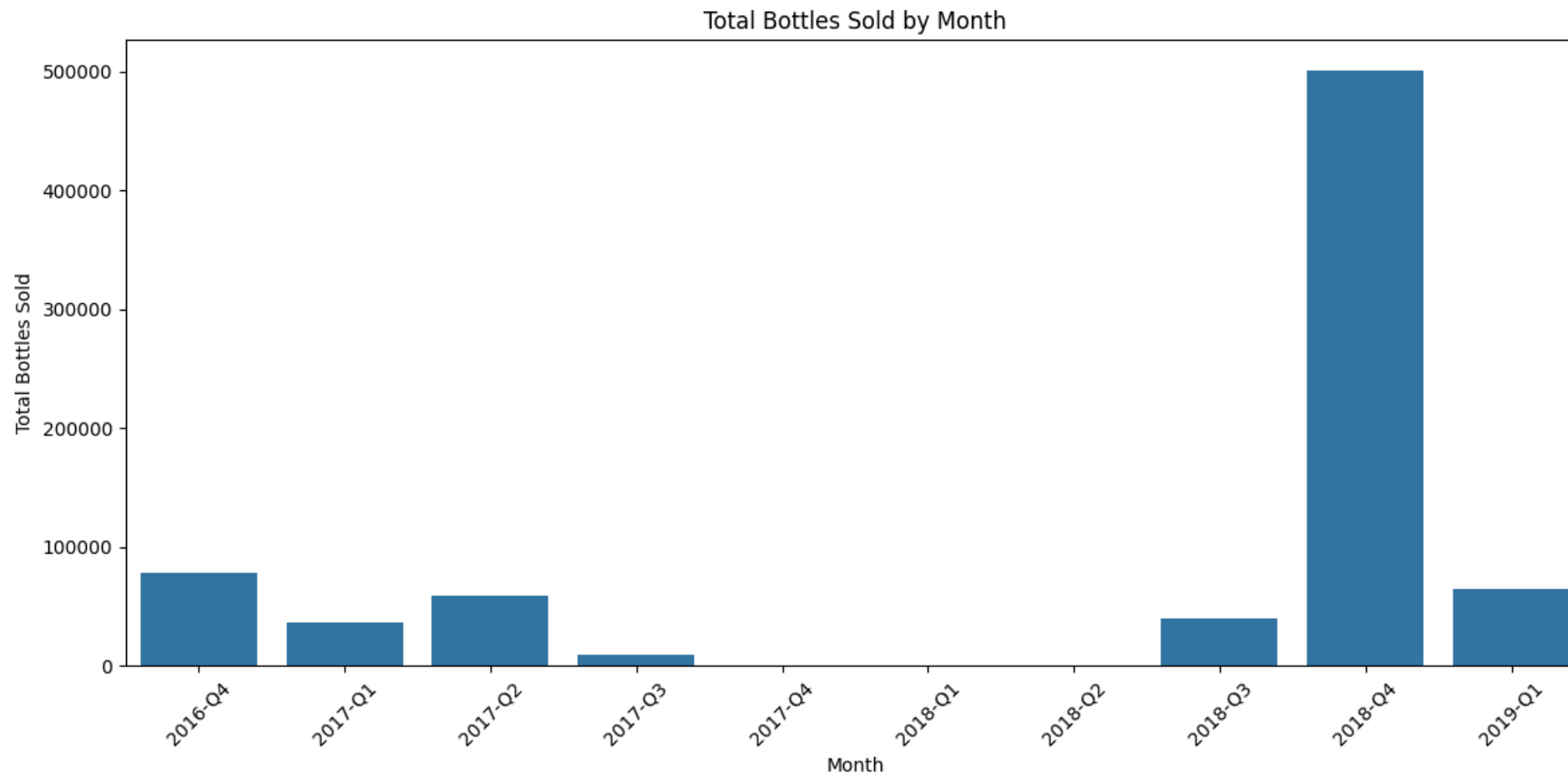
❖ What's next?





Data Loading

❖ What's next?



Review

Overview

1. Data-driven decision making
2. Business Intelligent
3. Data Architecture
4. ETL process overview

Section 1

Data Transformation

1. Transformation Problem
2. Cleaning
3. Processing

Section 3

Data Extraction

1. Extraction Process
2. Dataset: Iowa Liquor Sale
3. Case study: Extract Iowa Liquor Sale Dataset

Section 2

Data Loading

1. Dims, Fact
2. Slowly Changing Dimension
3. Full Pipeline

Section 4

